Event and waveform supplement: expanded

Independent archive reanalysis

13 October 2026

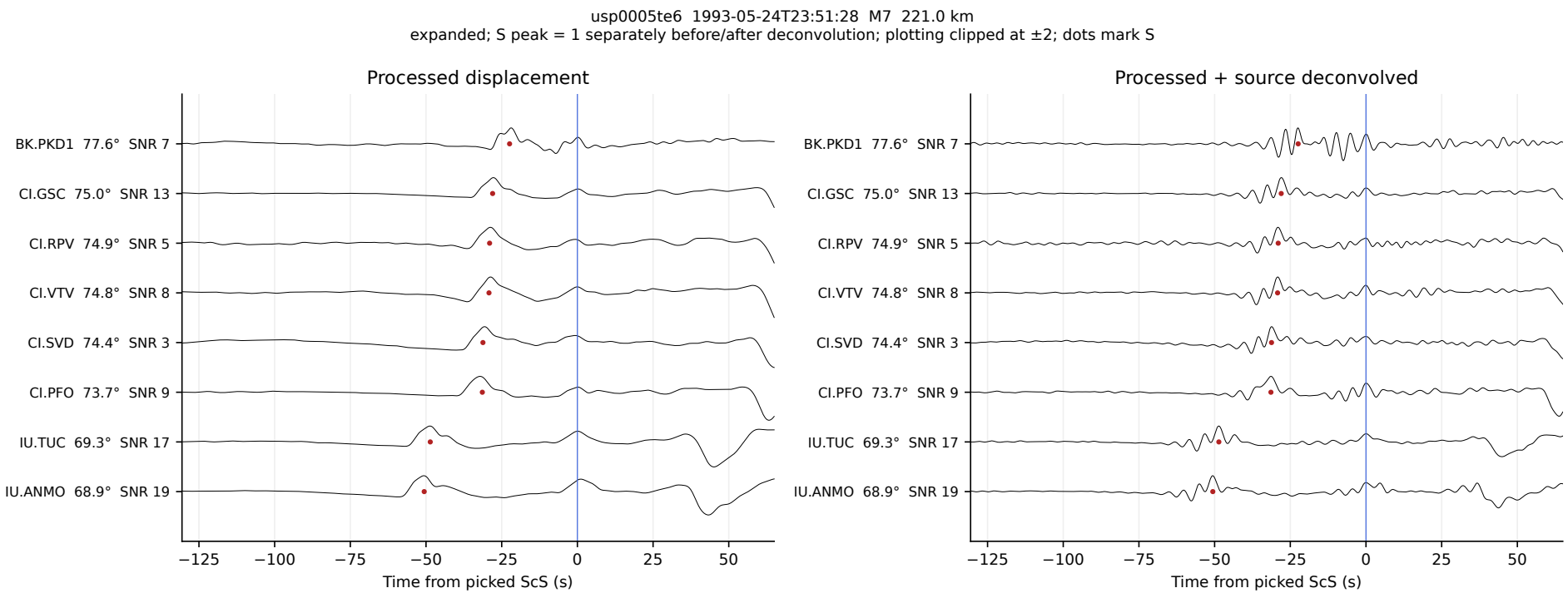
Each event below is a USGS catalog record. Counts refer to traces actually used in this dataset's images, not all downloaded stations. An asterisk flags a historical catalog value outside the published magnitude/depth bounds. Both processed displacement and processed plus source-deconvolved waveforms are shown for every used trace. Event pages with zero used traces give rejection summaries. Records retain at least 80 s before S and 65 s after ScS so noise can be assessed. The blue vertical line marks ScS; red dots mark S. The displacement and deconvolved curves are each normalized to their own S peak. Display clipping at twice that amplitude does not alter stored data. SNR is a peak-to-pre-S-RMS ratio. Receiver splitting corrections have not been applied because the complete original station/path correction set was not recovered. Phase selection is automatic and requires expert review.

USGS ID	UTC date/time	Latitude	Longitude	Depth km	Mag/type	Used
usp0004taw*	1991-06-23 21:22:28	-26.802	-63.349	558.0	7.3/mw	0
usp0005te6	1993-05-24 23:51:28	-22.671	-66.543	221.0	7.0/mwb	8
usp000628b	1993-10-19 04:02:21	-22.378	-65.971	272.0	6.0/mw	10
usp000669u	1994-01-10 15:53:50	-13.339	-69.446	596.0	6.9/mwb	0
usp0006bt7	1994-04-29 07:11:29	-28.299	-63.252	561.5	6.9/mwb	1
usp0006cbq	1994-05-10 06:36:28	-28.501	-63.096	600.5	6.9/mwb	0
usp0006h5m	1994-08-19 10:02:51	-26.642	-63.421	563.6	6.5/mw	2
usp0007w7v	1997-01-23 02:15:22	-21.999	-65.719	276.2	7.1/mwc	10
usp000853a	1997-07-20 10:14:22	-22.982	-66.301	256.1	6.1/mwc	0
usp0008bg2	1997-11-28 22:53:41	-13.740	-68.788	586.0	6.7/mwc	23
usp0008u3b*	1998-08-30 07:52:25	-23.934	-66.906	219.5	5.7/mwc	4
usp0009eex	1999-09-15 03:01:24	-20.934	-67.275	218.0	6.4/mwc	21
usp0009rrz	2000-04-23 17:01:17	-28.384	-62.943	609.8	6.1/mwc	1
usp0009suu	2000-05-12 18:43:18	-23.548	-66.452	225.0	7.2/mwc	0
usp000ahpv	2001-06-29 18:35:51	-19.522	-66.254	273.9	6.1/mwc	0
usp000bed7	2002-10-12 20:09:11	-8.295	-71.738	534.3	6.9/mwc	0
usp000bw3r	2003-04-27 22:57:44	-8.195	-71.592	559.9	6.0/mwb	0
usp000c0ag	2003-06-20 06:19:38	-7.606	-71.722	558.1	7.1/mwc	0
usp000c3gh	2003-07-27 11:41:27	-20.134	-65.185	345.3	6.0/mwb	57
usp000cq4u	2004-03-17 03:21:07	-21.118	-65.586	289.8	6.1/mwb	0
usp000d88b	2004-11-12 06:36:16	-26.705	-63.319	568.7	6.1/mwb	2
usp000djqf	2005-03-21 12:23:54	-24.983	-63.470	579.1	6.9/mwb	19
usp000djqg	2005-03-21 12:43:12	-24.725	-63.507	570.1	6.4/mwc	4
usp000etj9	2006-09-22 02:32:25	-26.868	-63.149	598.3	6.0/mwb	9
usp000ex8r	2006-11-13 01:26:35	-26.052	-63.283	572.0	6.8/mwb	9
usp000fgt2	2007-07-21 15:34:52	-22.151	-65.777	289.5	6.4/mwc	113
usp000fsu0	2007-11-18 05:40:12	-22.643	-66.323	246.4	6.0/mwb	26

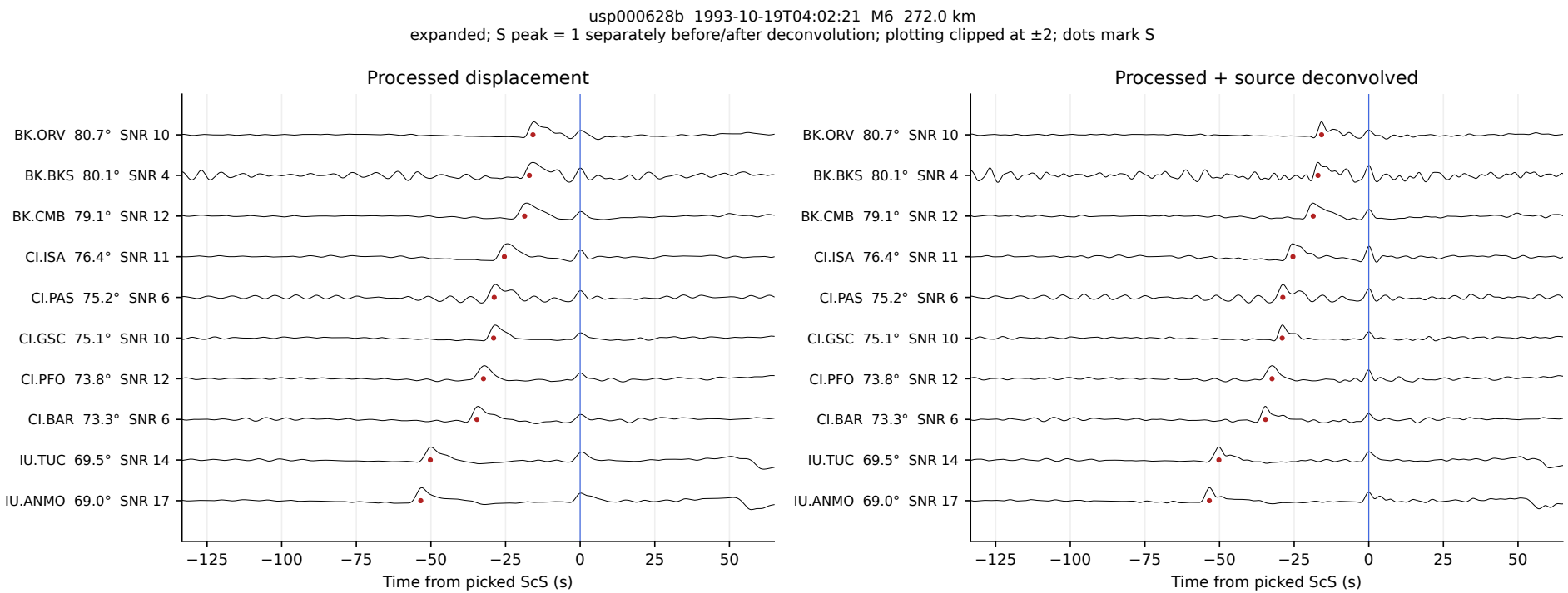
USGS ID	UTC date/time	Latitude	Longitude	Depth km	Mag/type	Used
usp000ggap	2008-09-03 11:25:14	-26.736	-63.225	569.6	6.3/mwc	48
usp000gk3x	2008-10-12 20:55:41	-20.123	-64.971	352.7	6.2/mwb	130
usp000h41w	2009-11-14 19:44:29	-22.965	-66.641	220.4	6.2/mwc	27
usp000hd4k	2010-05-24 16:18:29	-8.087	-71.558	581.2	6.5/mwc	0
usp000hsdc	2011-01-01 09:56:58	-26.803	-63.136	576.8	7.0/mww	7
usp000j7ee	2011-09-02 13:47:09	-28.398	-63.029	578.9	6.7/mww	0
usp000jb98	2011-11-22 18:48:16	-15.364	-65.090	549.9	6.6/mww	34
usp000jftj	2012-03-05 07:46:10	-28.246	-63.294	553.9	6.1/mww	0
usp000jm15	2012-05-28 05:07:23	-28.043	-63.094	586.9	6.7/mww	1
usc000fbl2	2013-02-22 12:01:58	-27.932	-63.097	575.2	6.1/mww	6
usb000sfj5	2014-09-24 11:16:15	-23.801	-66.632	224.0	6.2/mww	0
usc000tp1s	2015-02-11 18:57:22	-23.113	-66.688	223.0	6.7/mww	0
us100041f1	2015-11-26 05:45:18	-9.182	-71.257	602.8	6.7/mww	0
us10004ebw	2016-01-14 03:25:28	-19.760	-63.329	582.6	6.1/mww	25
us10006a1d	2016-08-04 14:15:12	-22.334	-66.008	270.0	6.2/mww	49
us20008kn8	2017-02-18 12:10:17	-23.861	-66.659	222.0	6.4/mww	13
us20008l41	2017-02-21 14:09:04	-19.281	-63.905	596.0	6.5/mww	52
us1000dce2	2018-04-02 13:40:34	-20.659	-63.006	559.0	6.8/mww	0
us2000j0ja	2019-01-05 19:25:38	-8.144	-71.587	570.4	6.8/mww	0
us1000j96d	2019-03-01 08:50:42	-14.713	-70.155	267.0	7.0/mww	75
us1000jg5z	2019-03-15 05:03:50	-17.874	-65.907	359.0	6.3/mww	0
us70006qk4	2019-12-24 16:43:33	-26.964	-63.386	572.4	6.0/mww	2
us7000gmw3	2022-02-22 06:12:50	-22.666	-66.266	251.0	6.0/mww	20
us7000h8g3	2022-05-10 23:06:32	-23.496	-66.652	220.0	6.8/mww	63
us7000hcmn	2022-05-26 12:02:21	-14.908	-70.292	236.0	7.2/mww	0
us7000j5x2	2023-01-20 22:09:39	-26.751	-63.082	596.8	6.8/mww	0
us6000jiyj	2023-01-24 18:36:58	-26.720	-63.113	580.0	6.4/mww	8
us7000jlxu	2023-03-22 16:00:34	-23.422	-66.495	228.0	6.4/mww	0
us6000ky1y	2023-08-05 07:20:06	-28.177	-63.184	575.0	6.2/mww	2
us7000kqbl	2023-08-23 14:22:43	-26.884	-63.295	554.0	6.2/mww	8
us6000legn	2023-10-10 10:01:31	-22.915	-66.255	253.0	6.0/mww	0
us6000m52p	2024-01-20 21:31:05	-7.288	-71.464	607.0	6.6/mww	24

usp0004taw / 1991-06-23T21:22:28.940000Z
No accepted trace in this dataset.
phase peak at search boundary: 4
ScS peak at joint search boundary: 1

usp0005te6: 8 used traces; page 1/1

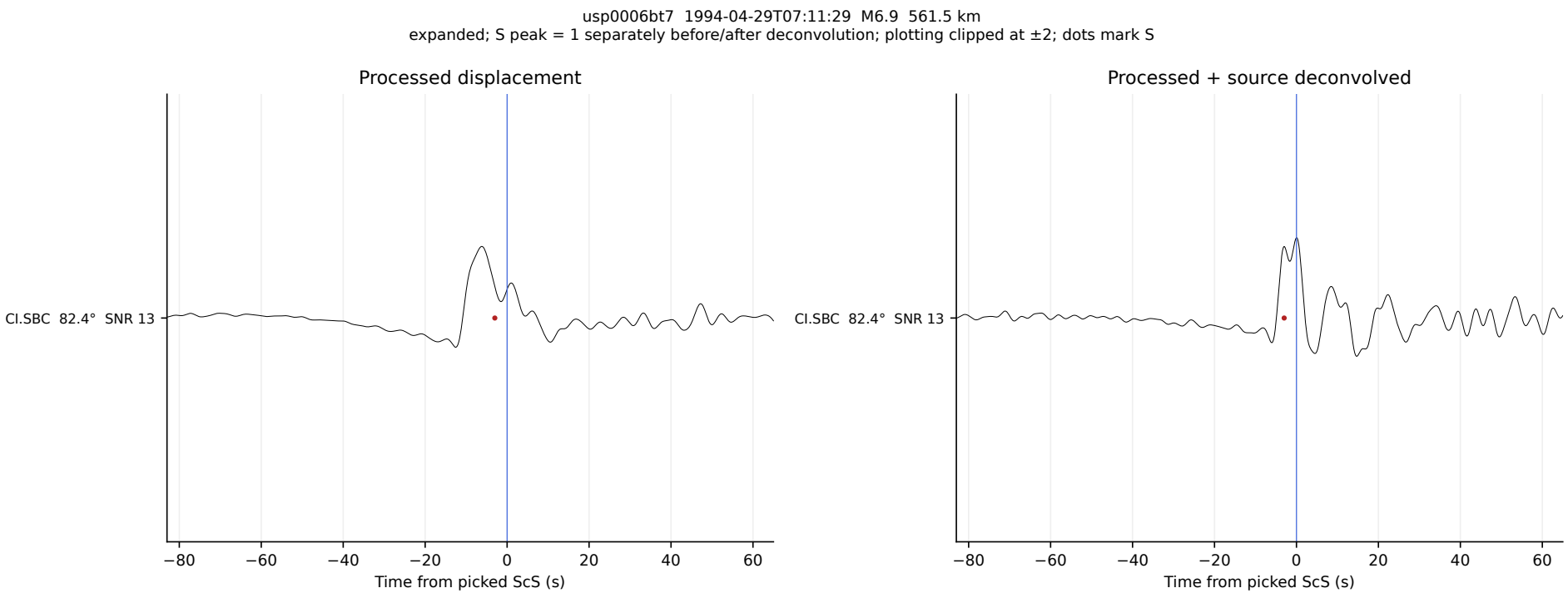


usp000628b: 10 used traces; page 1/1



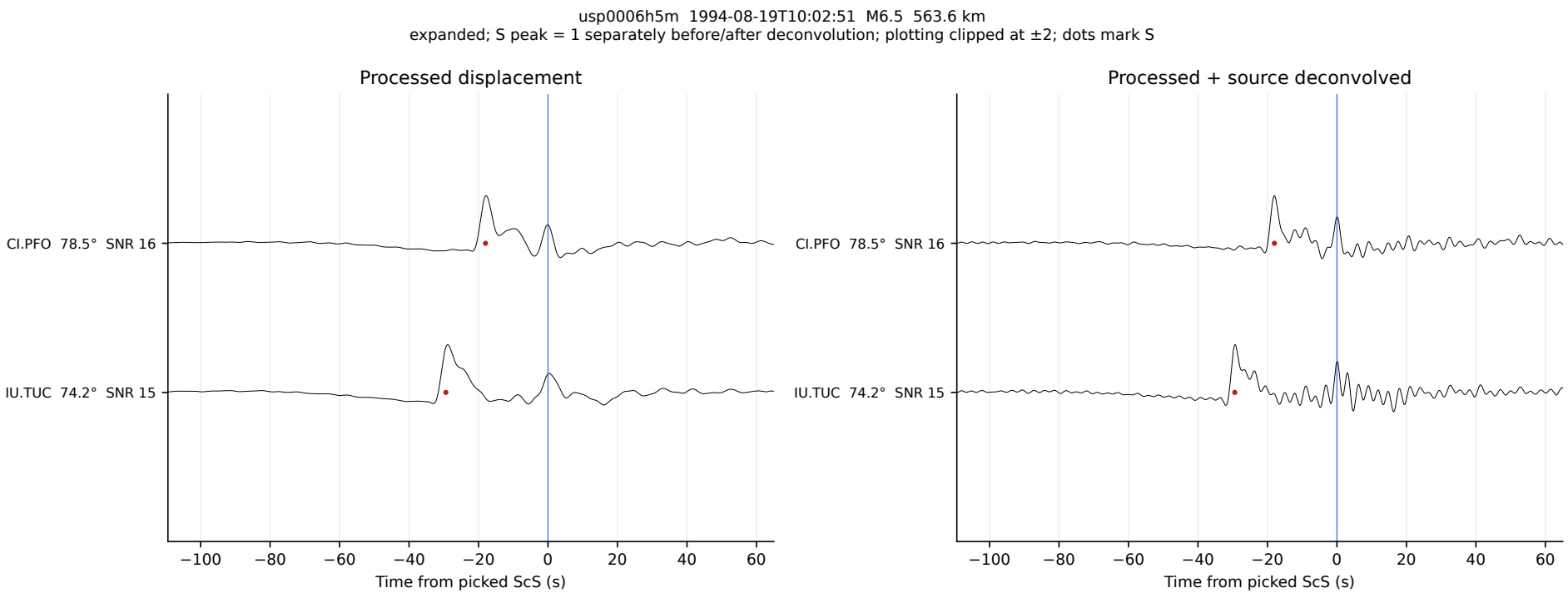
usp000669u / 1994-01-10T15:53:50.110000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 17

usp0006bt7: 1 used traces; page 1/1

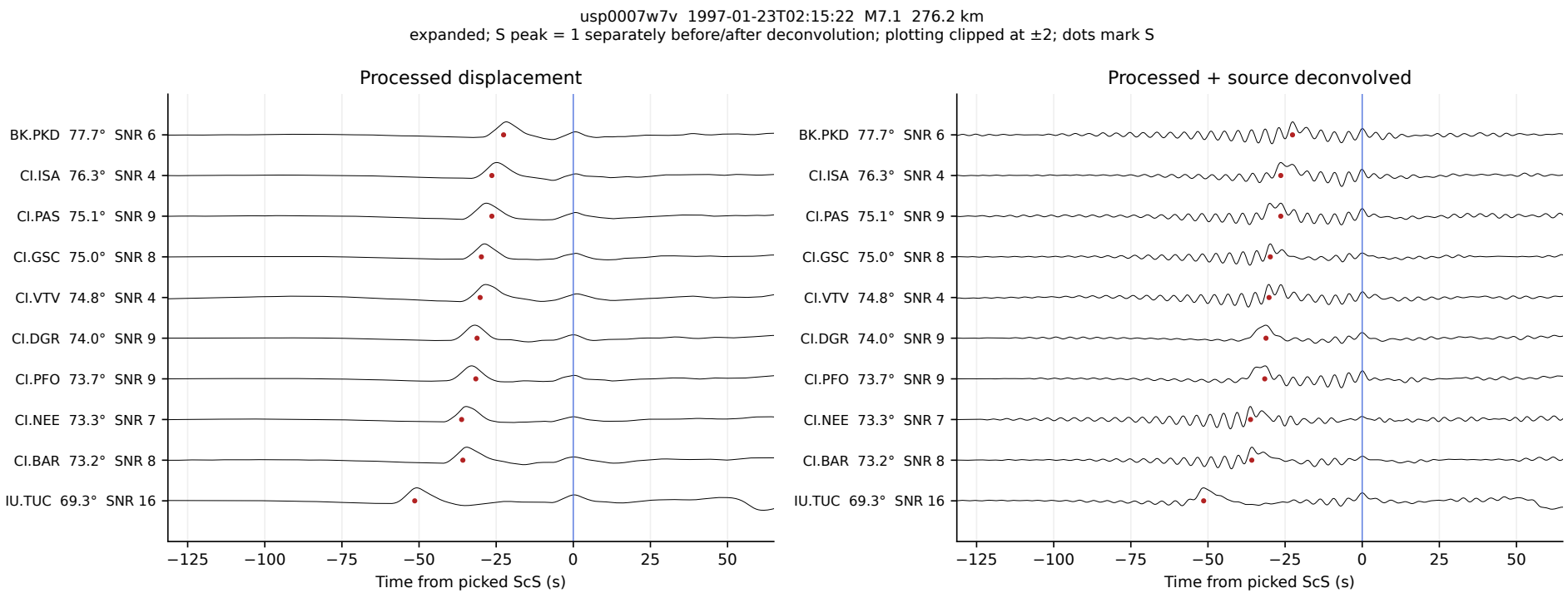


usp0006cbq / 1994-05-10T06:36:28.380000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 14
S or ScS below peak/RMS SNR threshold: 1

usp0006h5m: 2 used traces; page 1/1

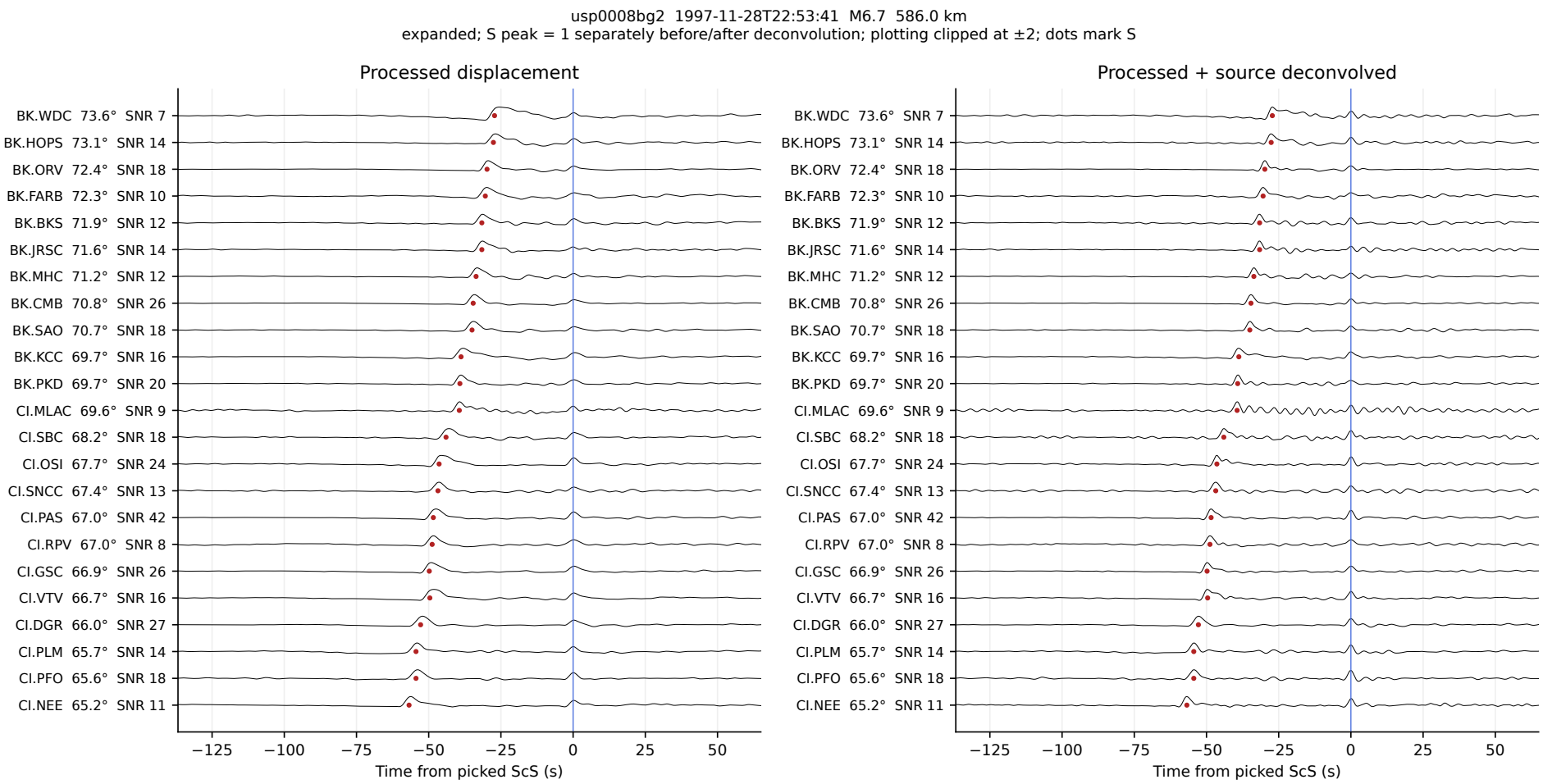


usp0007w7v: 10 used traces; page 1/1

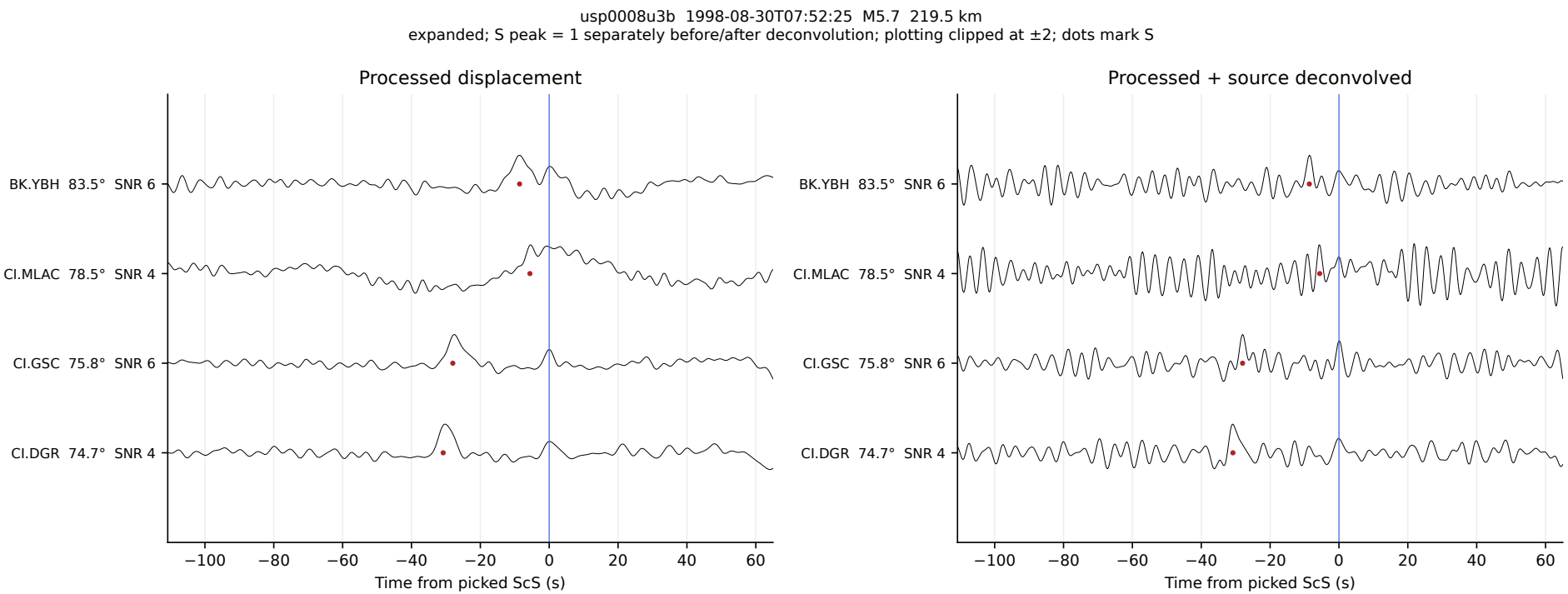


usp000853a / 1997-07-20T10:14:22.800000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 20

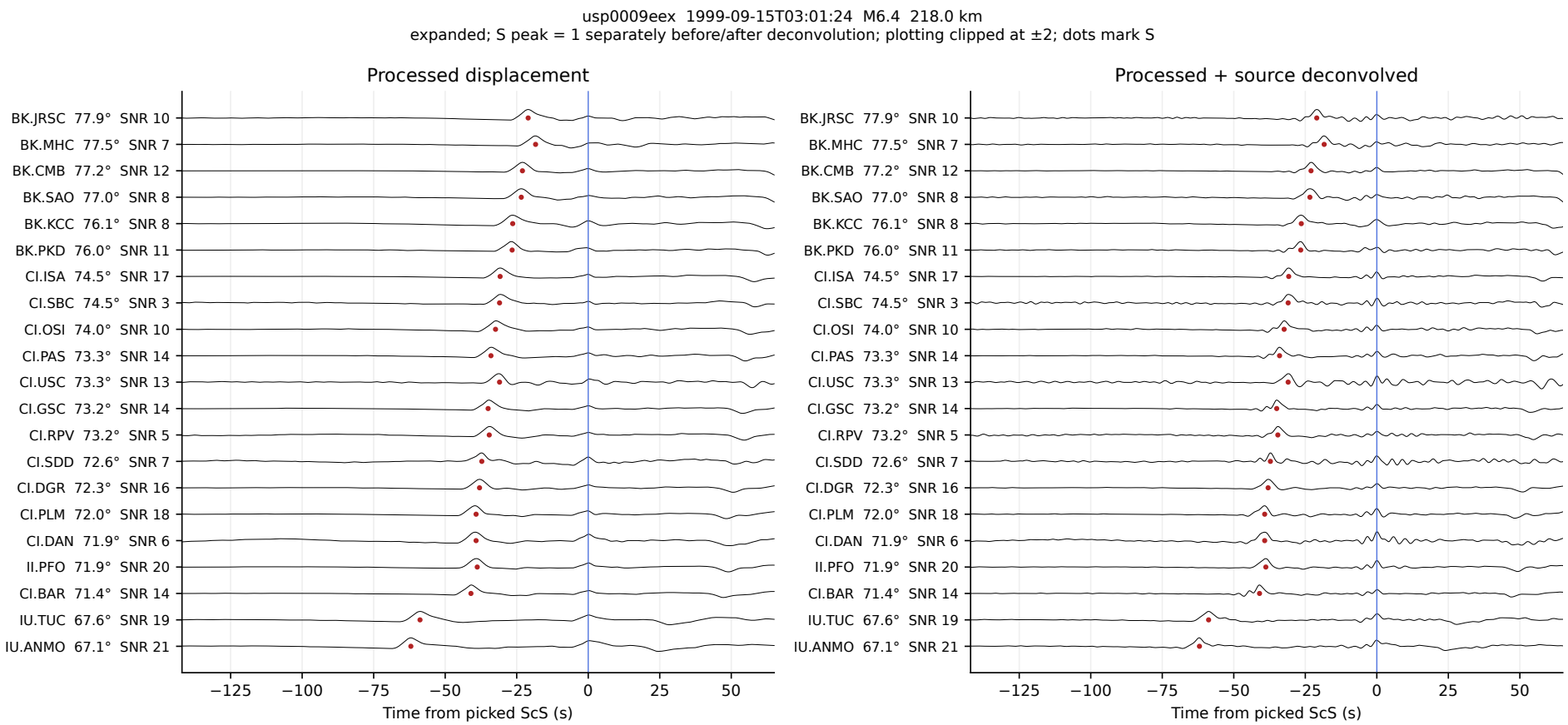
usp0008bg2: 23 used traces; page 1/1



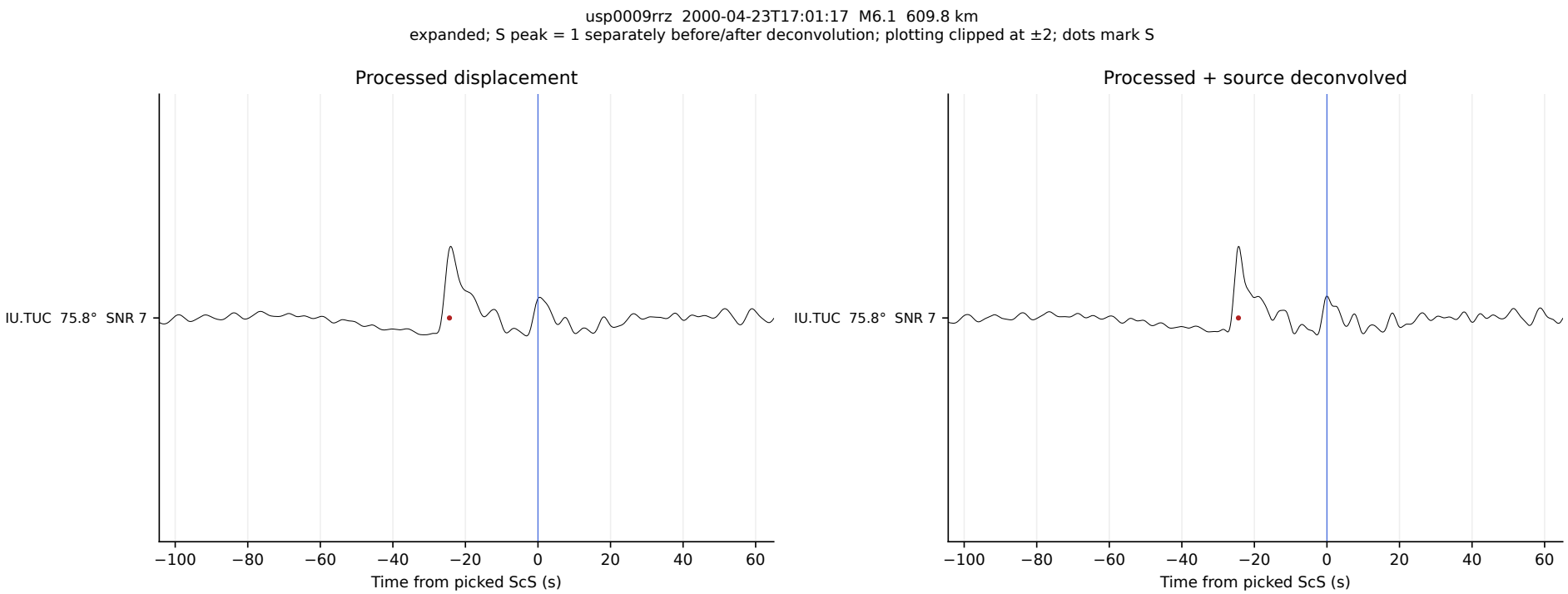
usp0008u3b: 4 used traces; page 1/1



usp0009eex: 21 used traces; page 1/1



usp0009rrz: 1 used traces; page 1/1



usp0009suu / 2000-05-12T18:43:18.120000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 53

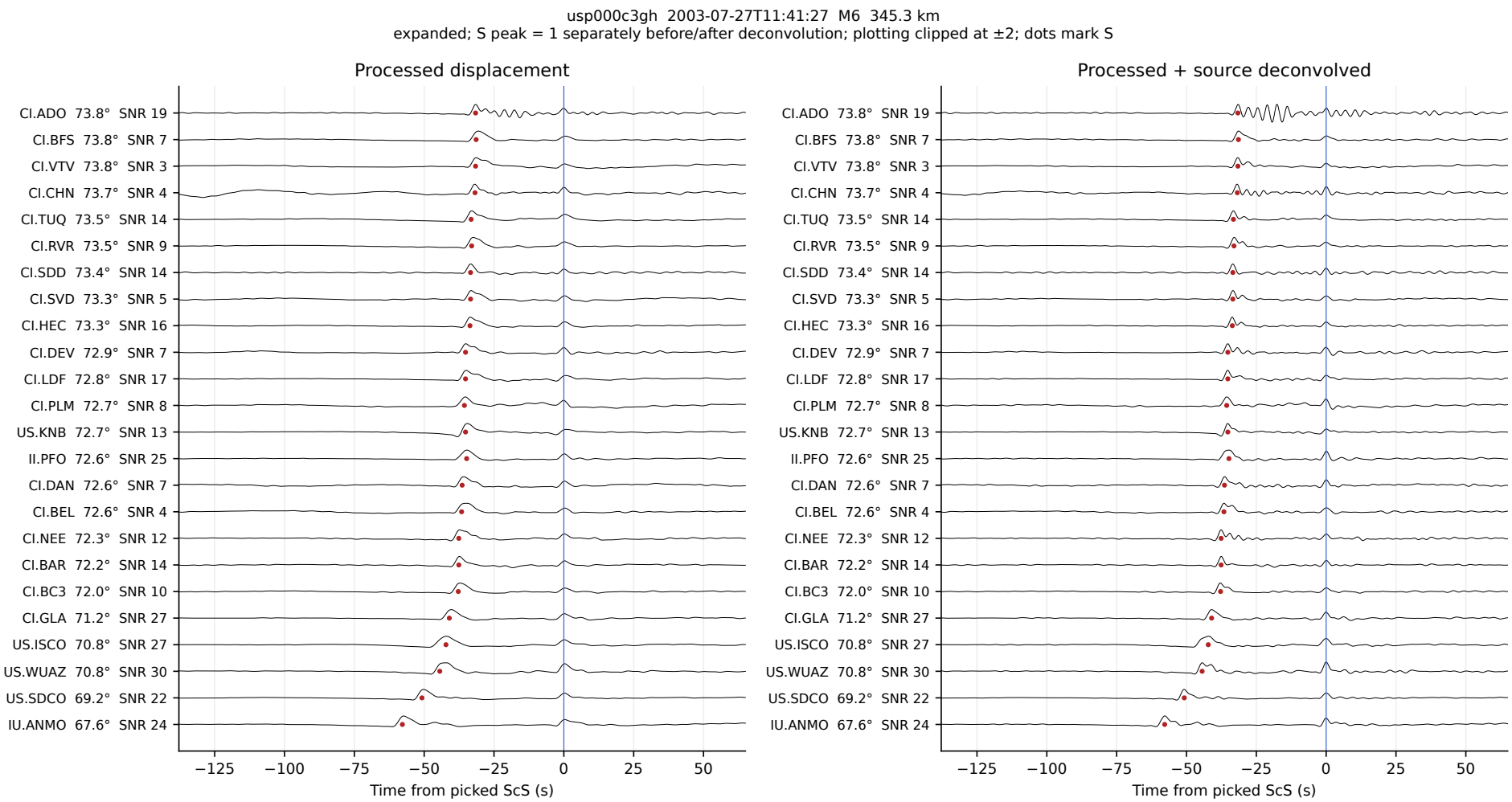
usp000ahpv / 2001-06-29T18:35:51.980000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 59

usp000bed7 / 2002-10-12T20:09:11.460000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 19

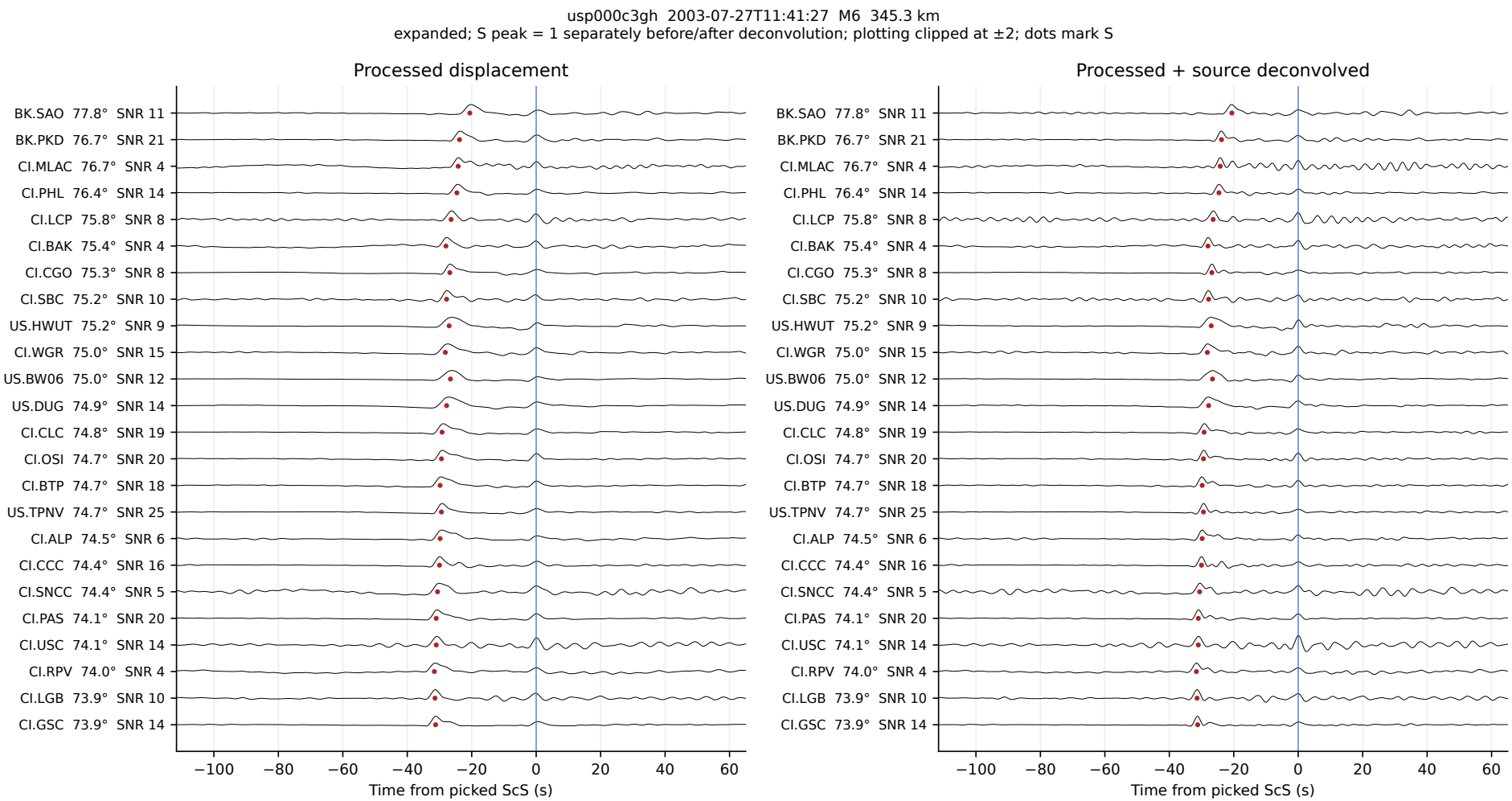
usp000bw3r / 2003-04-27T22:57:44.840000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 21

usp000c0ag / 2003-06-20T06:19:38.910000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 21

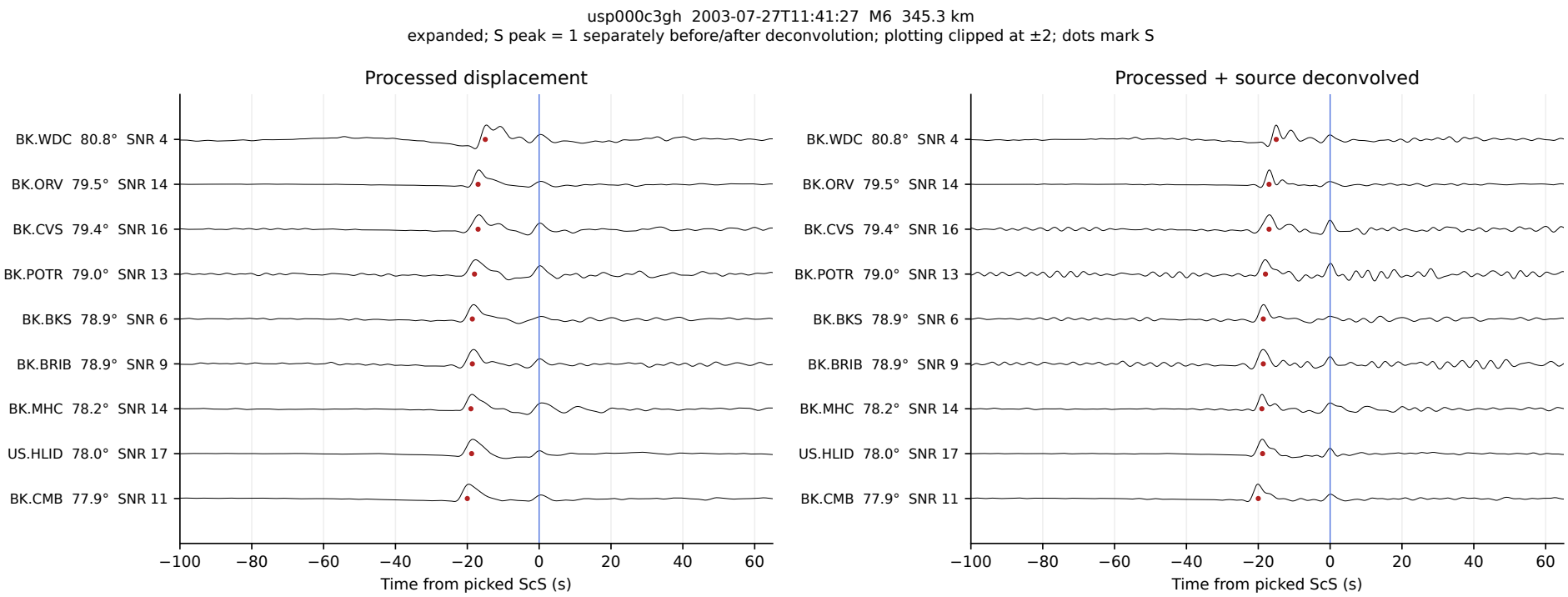
usp000c3gh: 57 used traces; page 1/3



usp000c3gh: 57 used traces; page 2/3

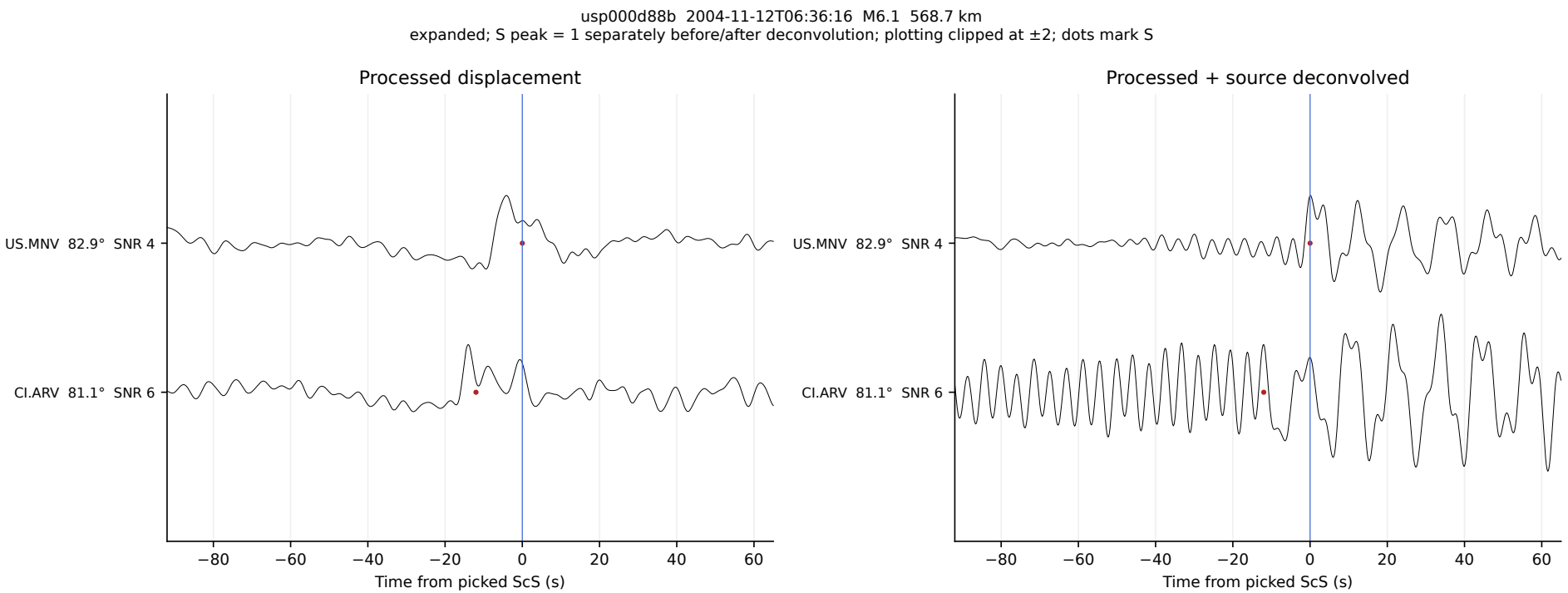


usp000c3gh: 57 used traces; page 3/3

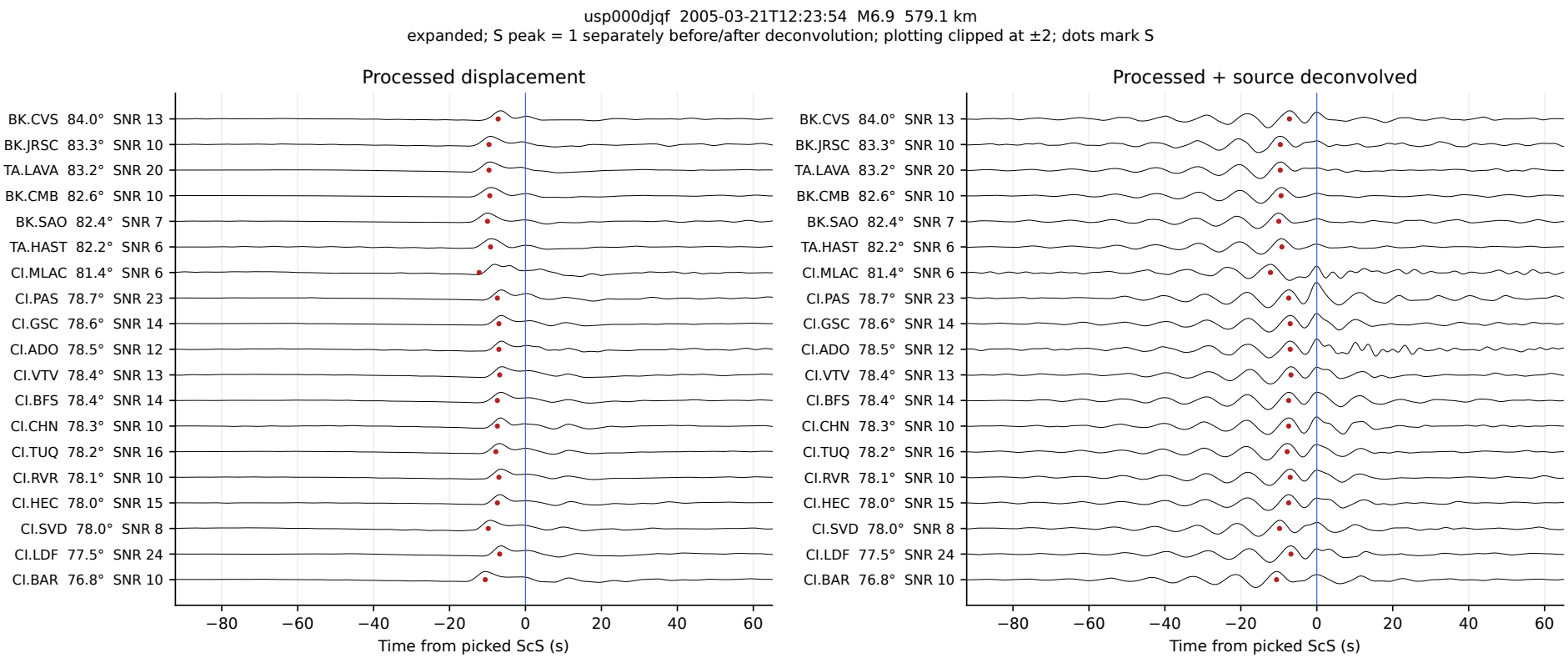


usp000cq4u / 2004-03-17T03:21:07.910000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 85

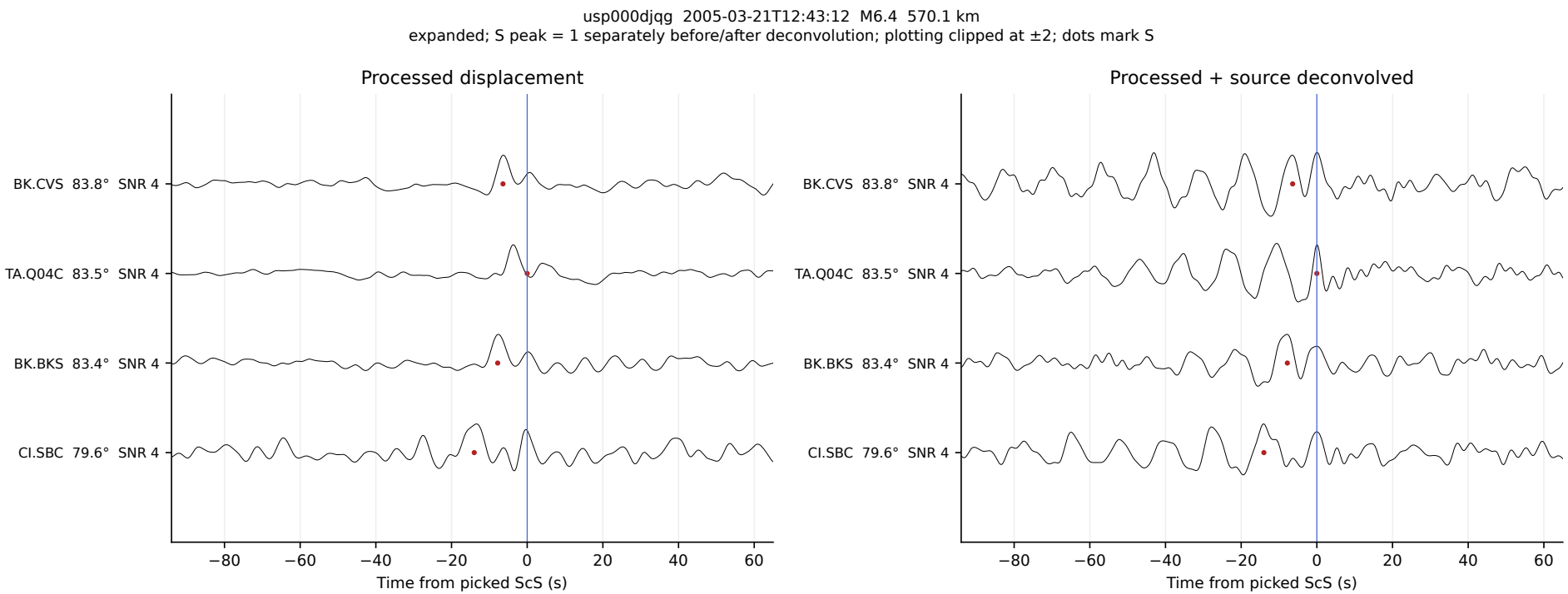
usp000d88b: 2 used traces; page 1/1



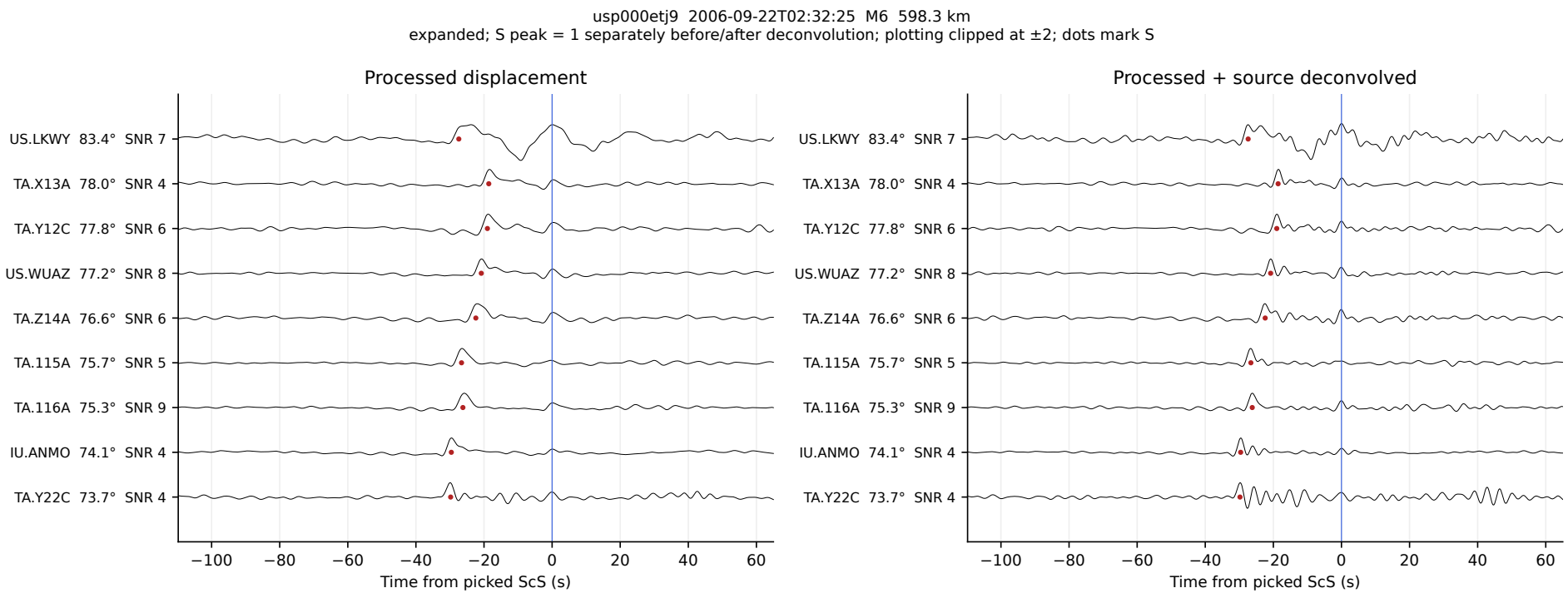
usp000djqf: 19 used traces; page 1/1



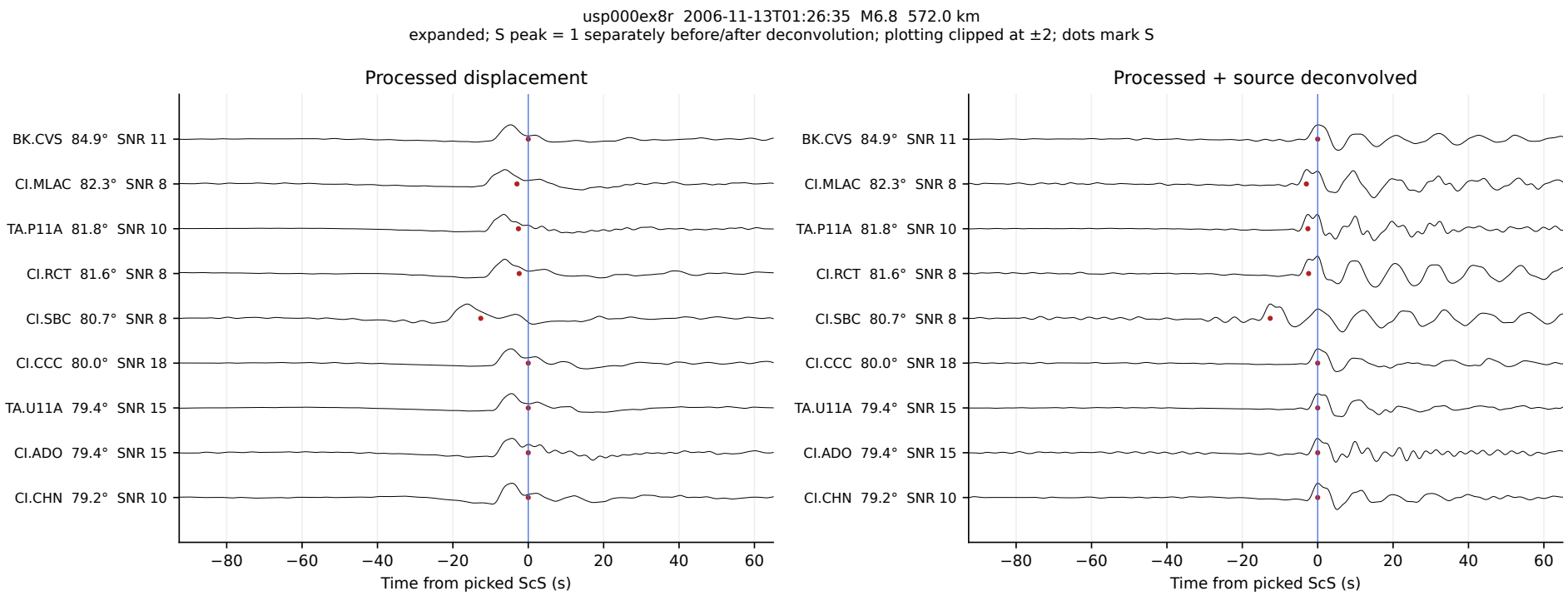
usp000djqg: 4 used traces; page 1/1



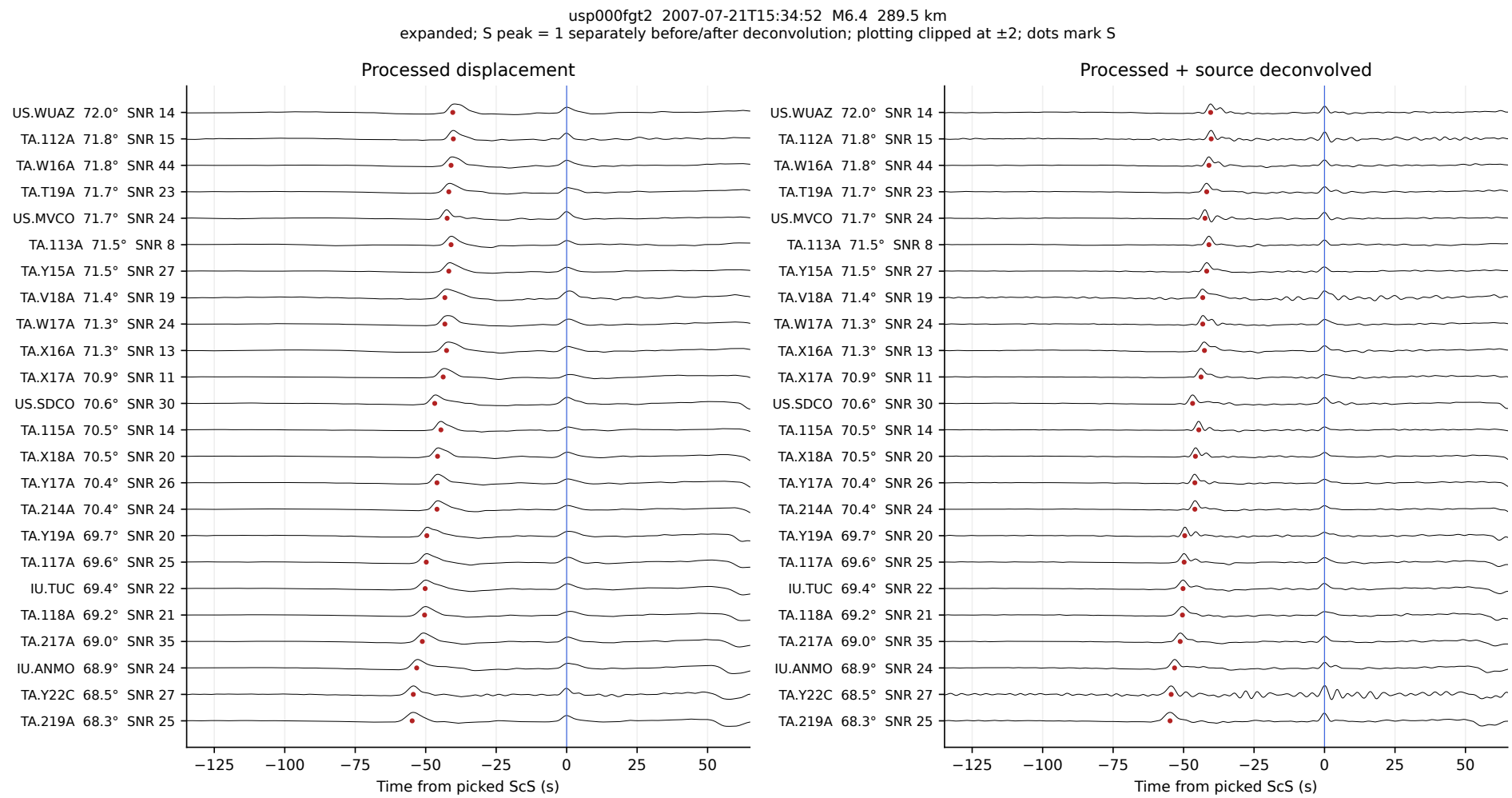
usp000etj9: 9 used traces; page 1/1



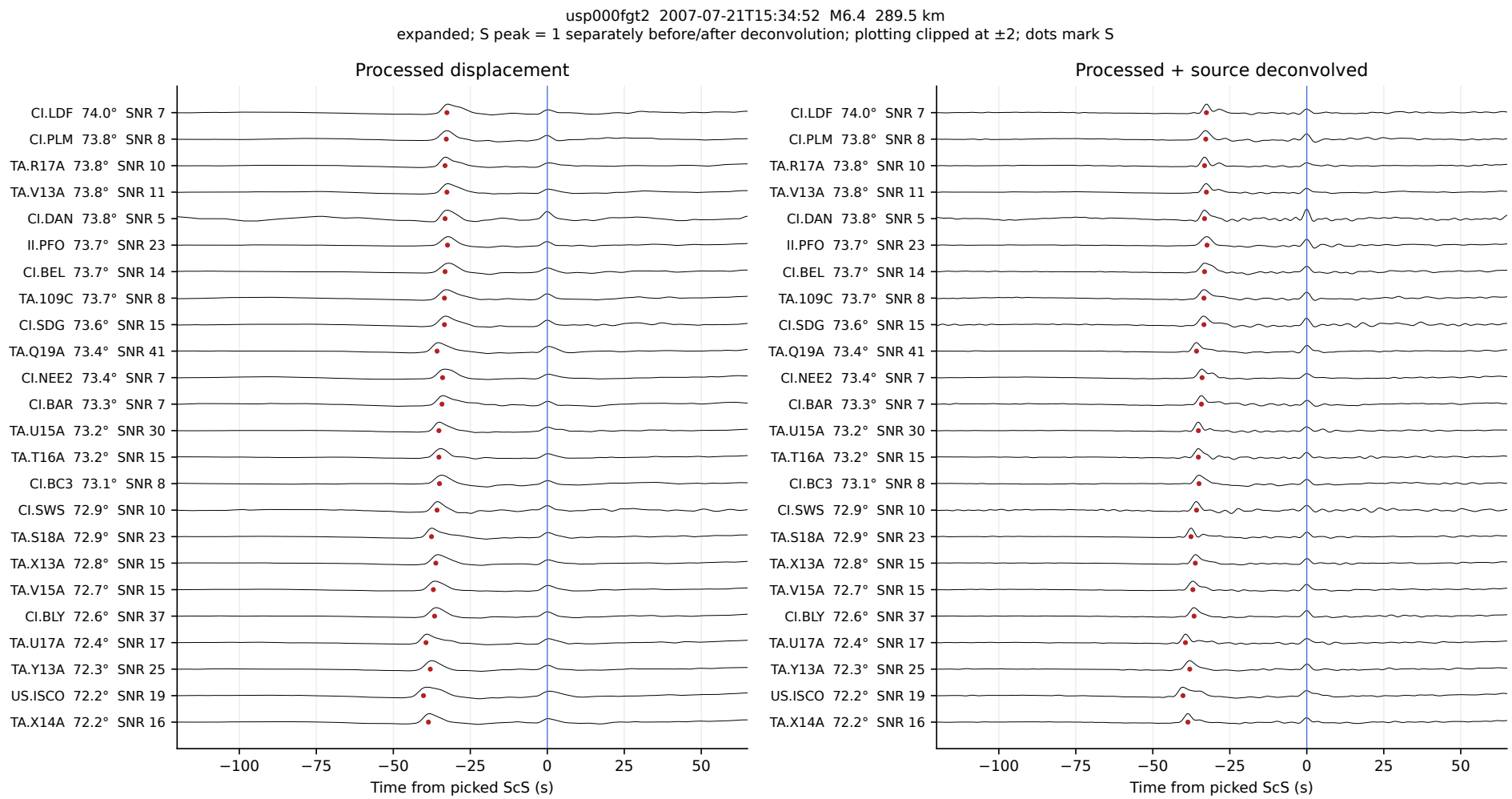
usp000ex8r: 9 used traces; page 1/1



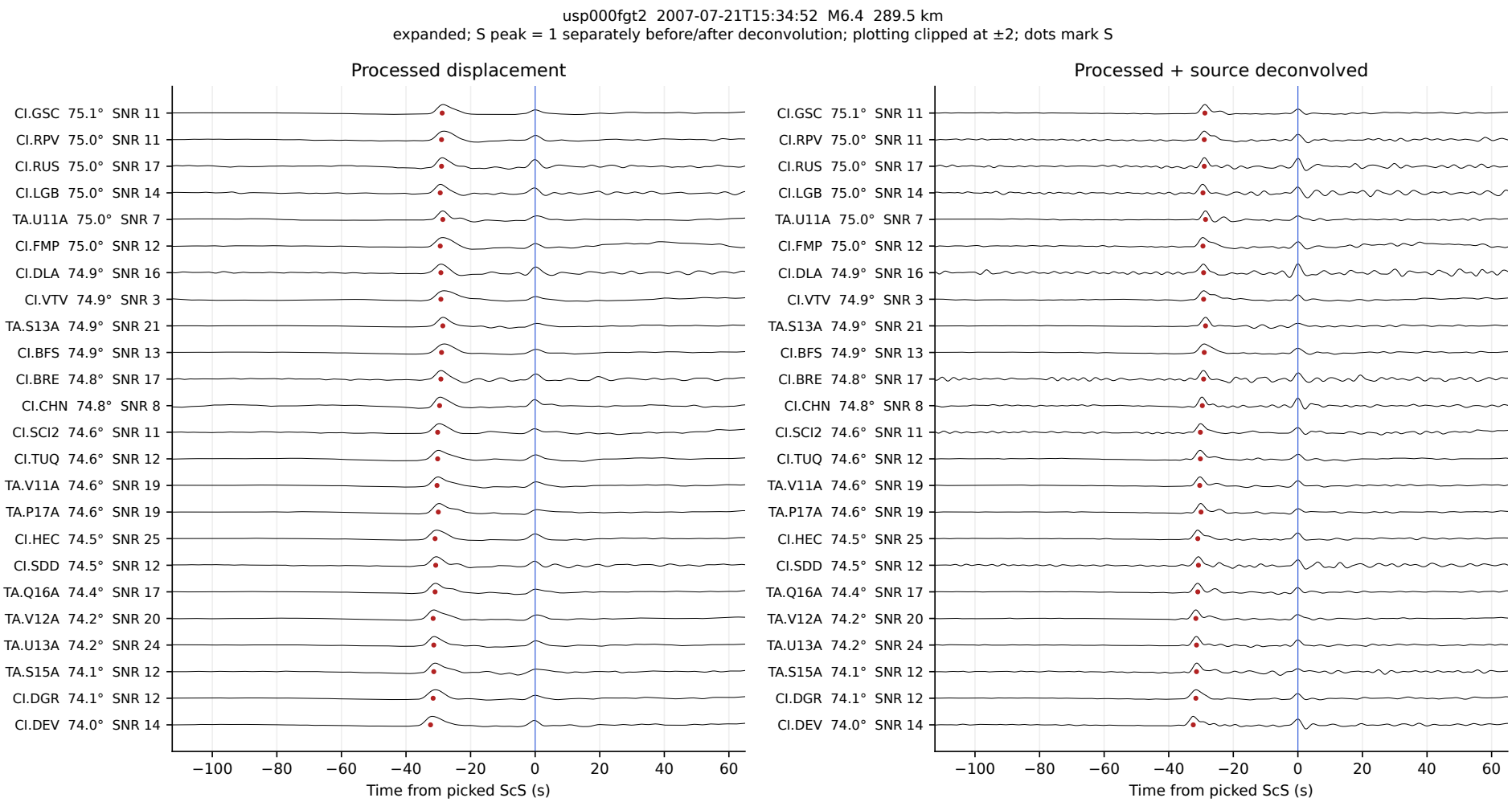
usp000fgt2: 113 used traces; page 1/5



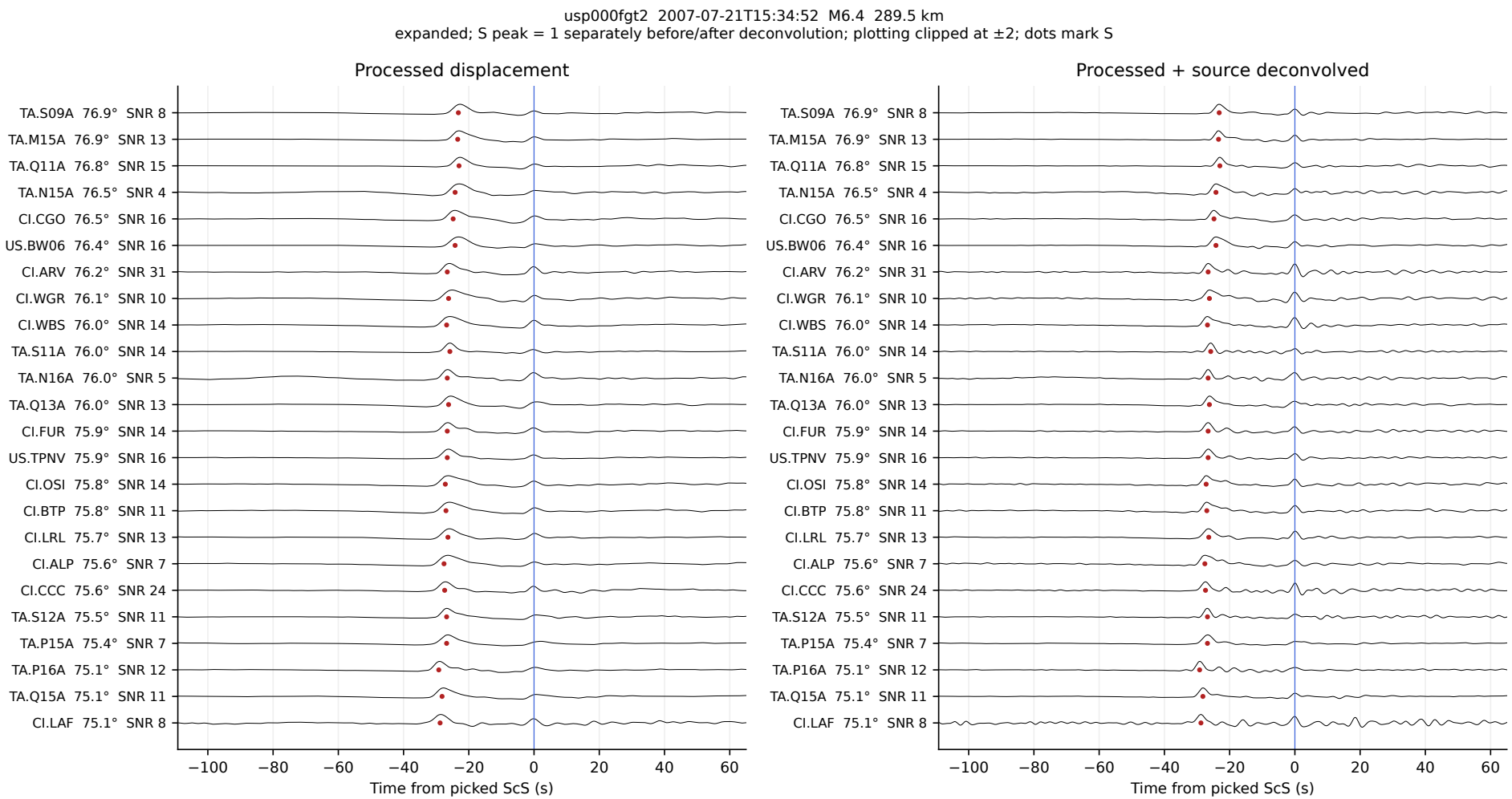
usp000fgt2: 113 used traces; page 2/5



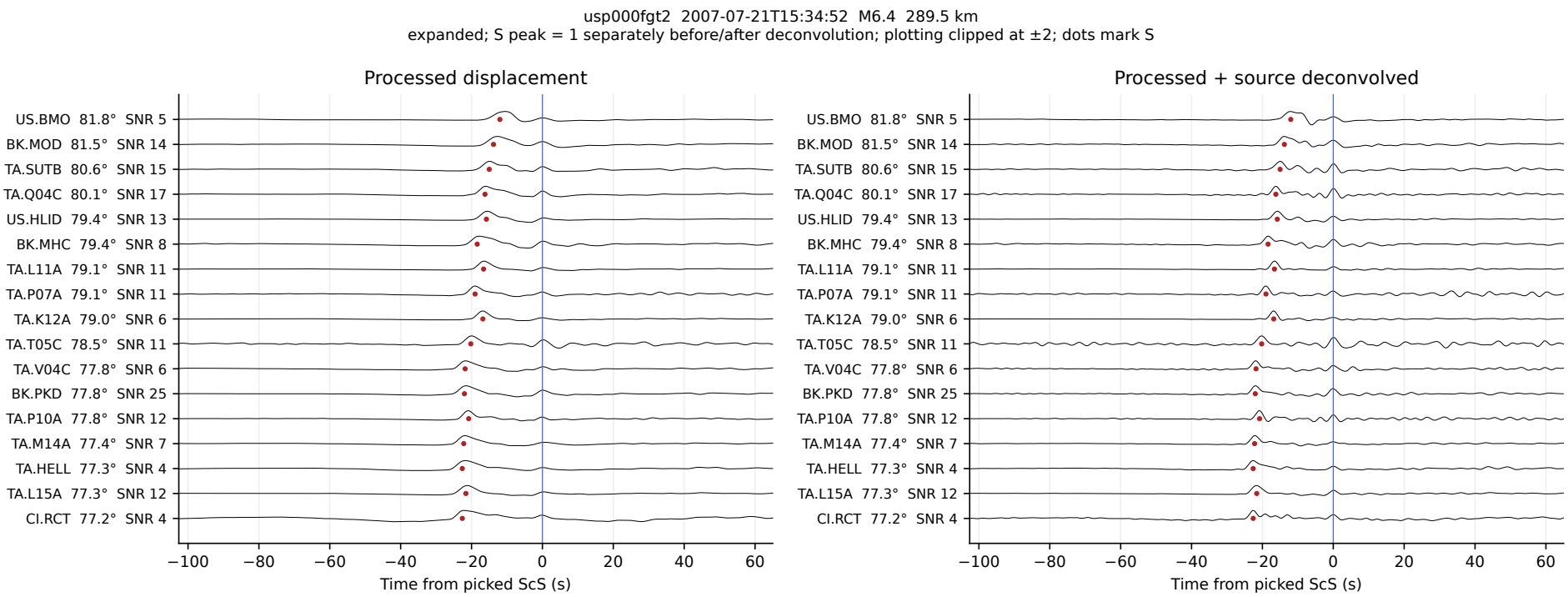
usp000fgt2: 113 used traces; page 3/5



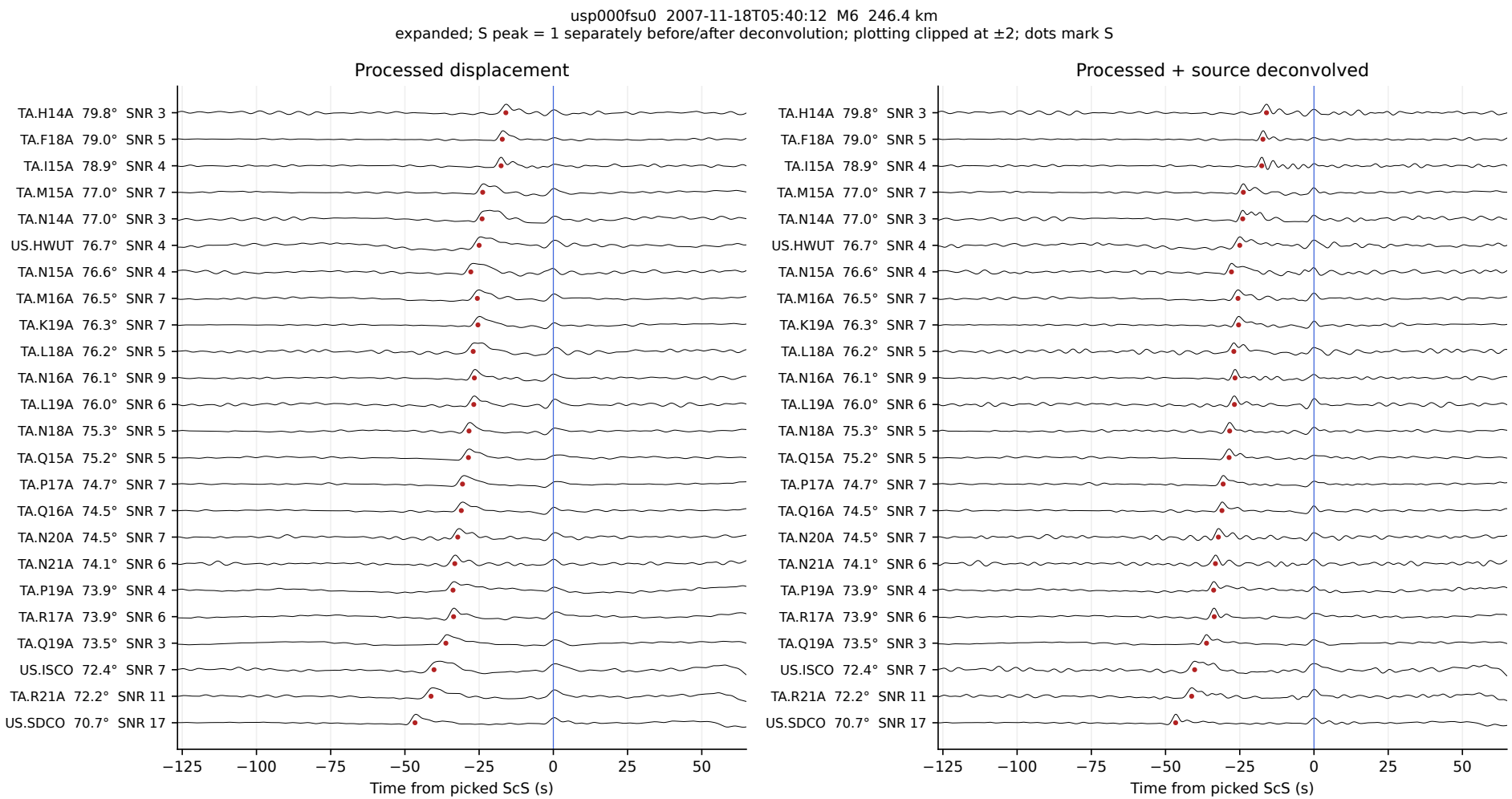
usp000fgt2: 113 used traces; page 4/5



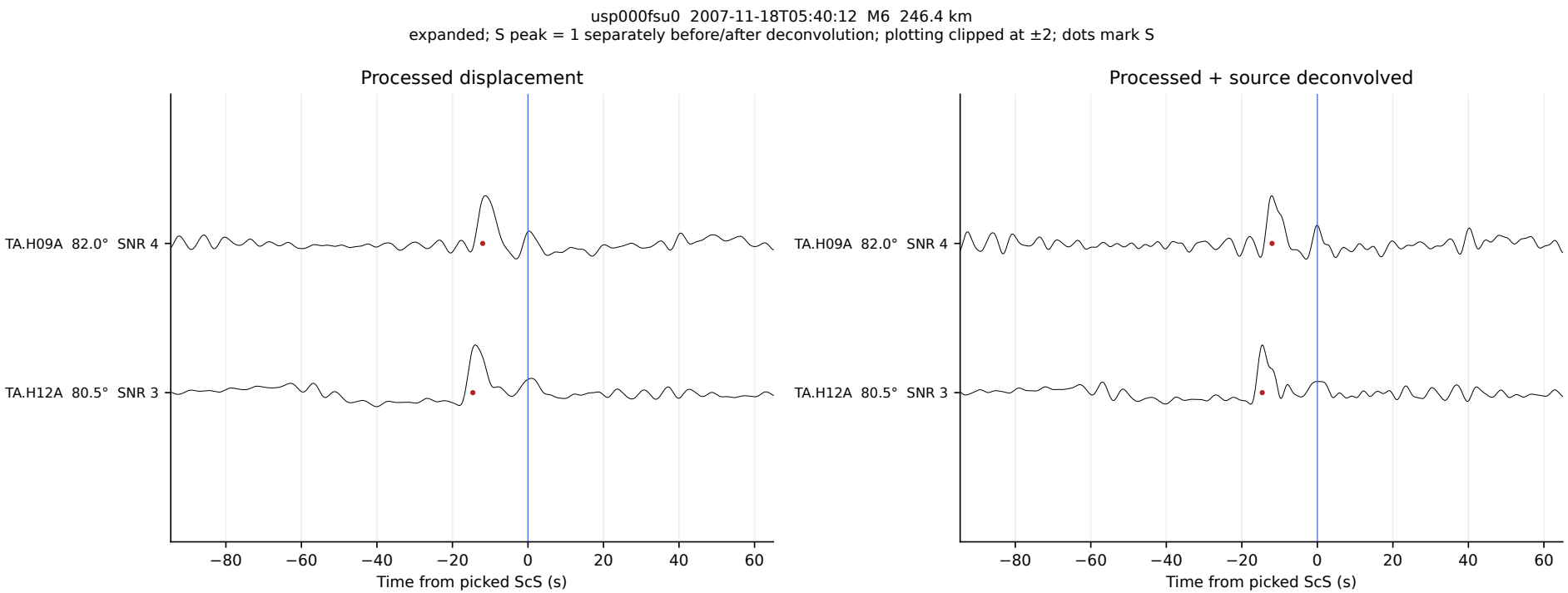
usp000fgt2: 113 used traces; page 5/5



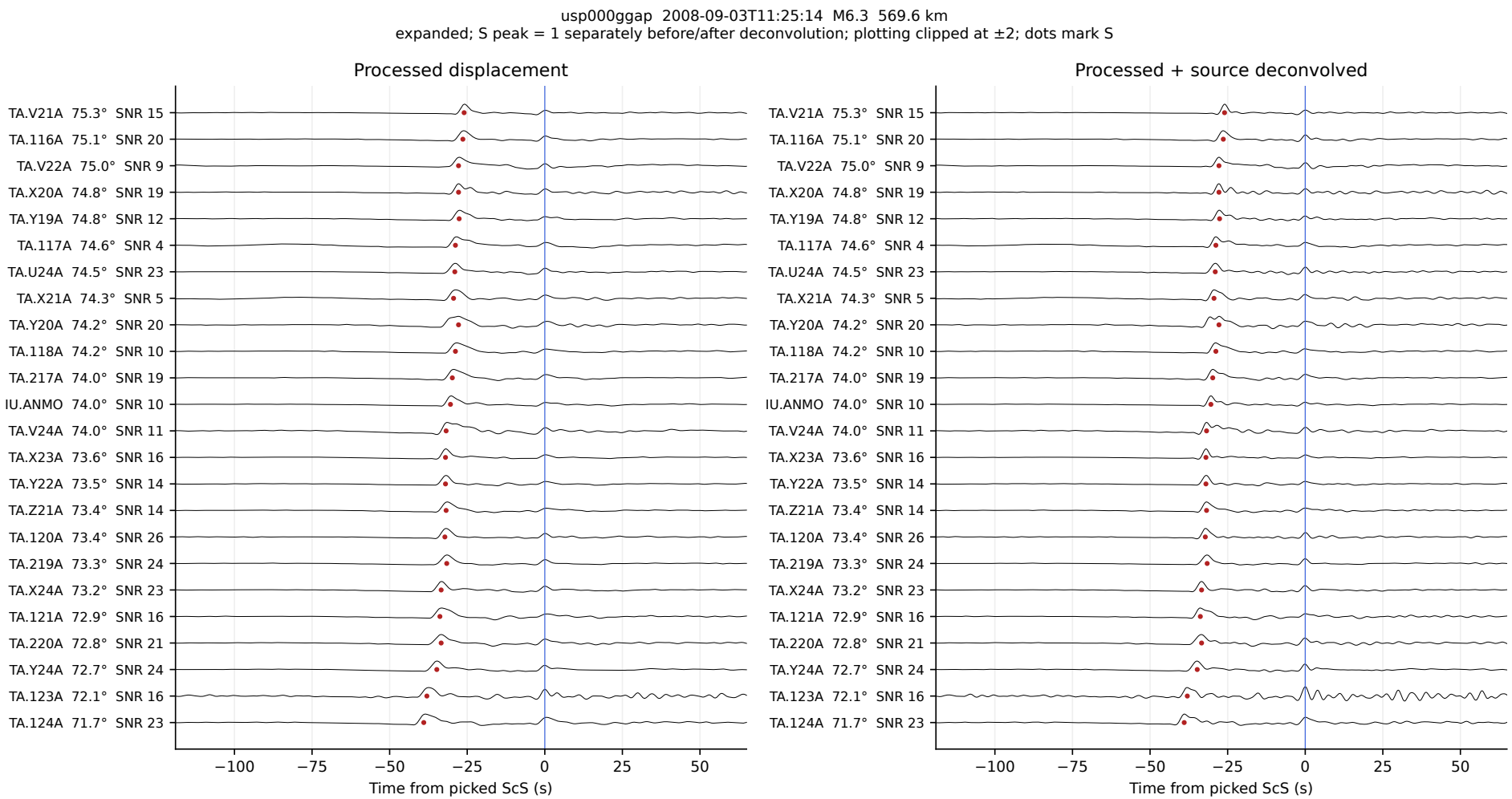
usp000fsu0: 26 used traces; page 1/2



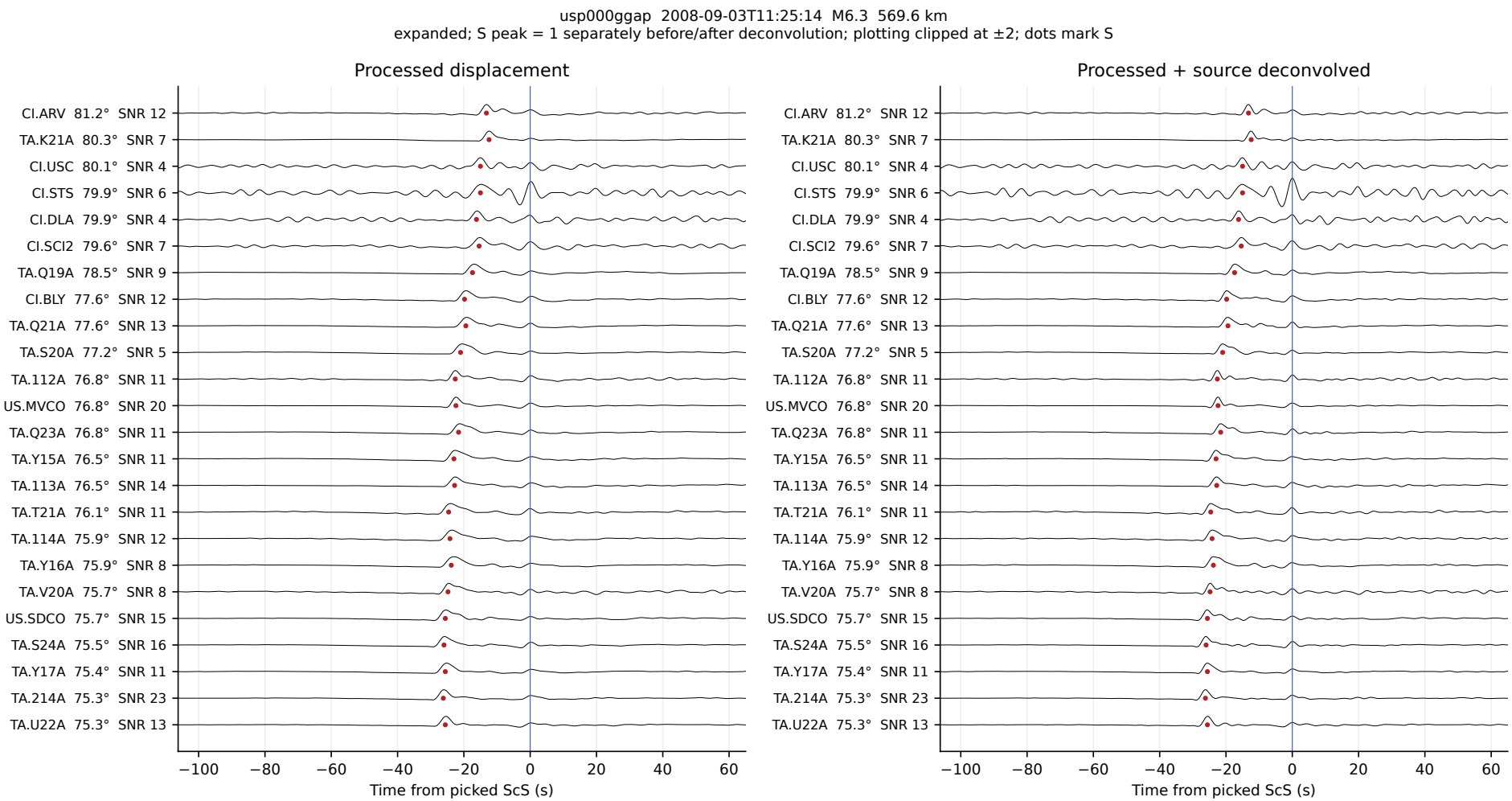
usp000fsu0: 26 used traces; page 2/2



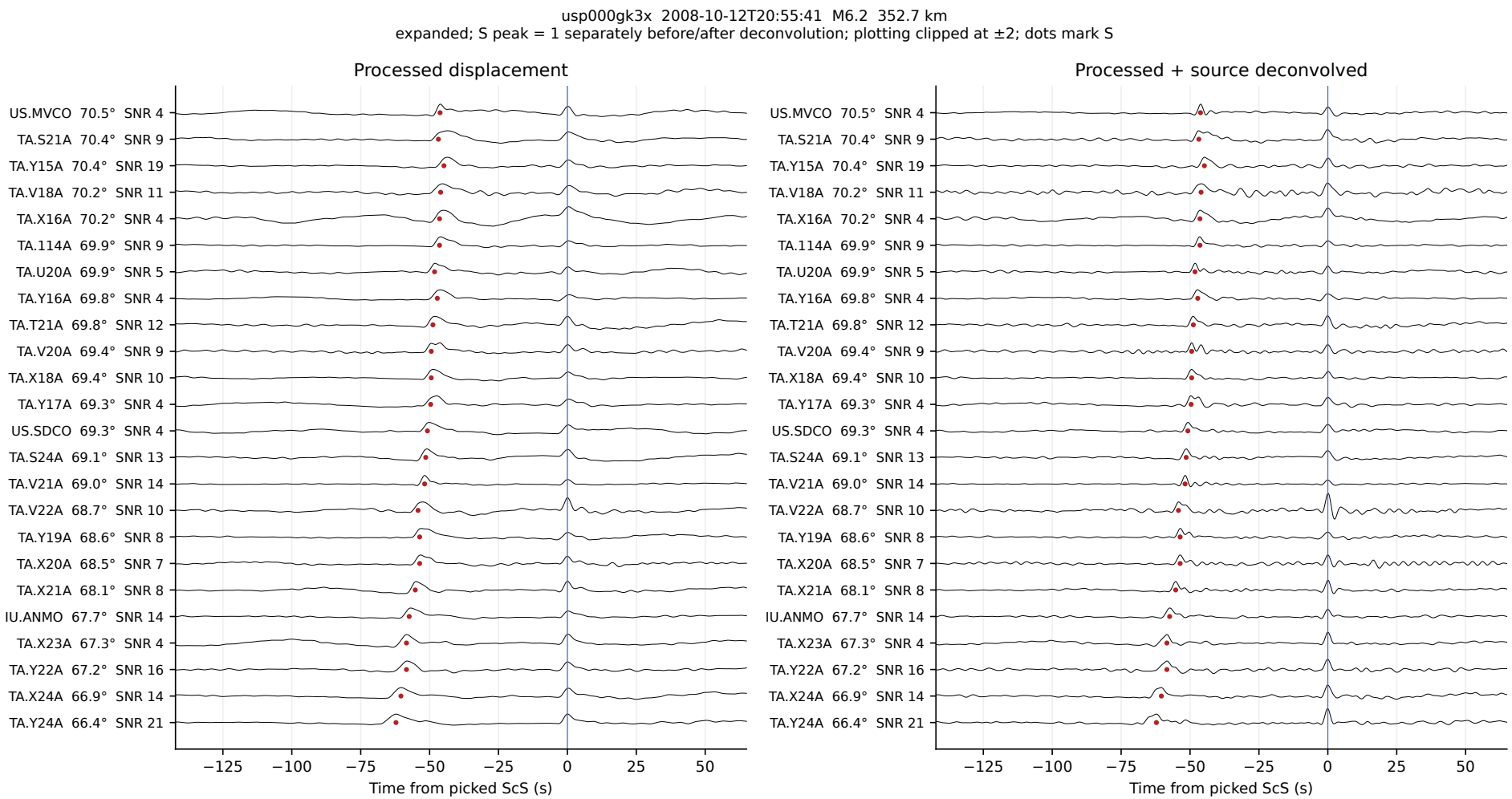
usp000ggap: 48 used traces; page 1/2



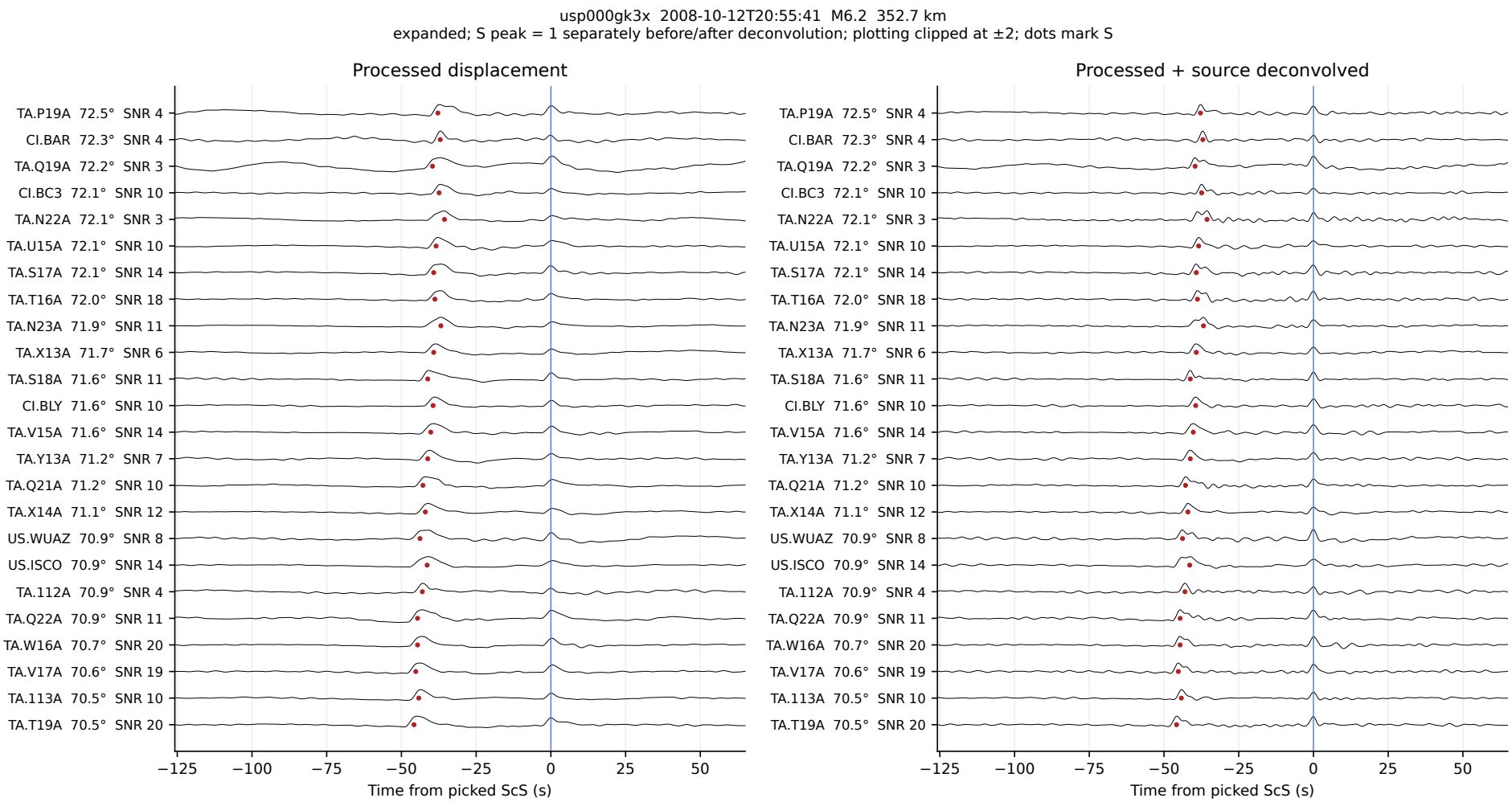
usp000ggap: 48 used traces; page 2/2



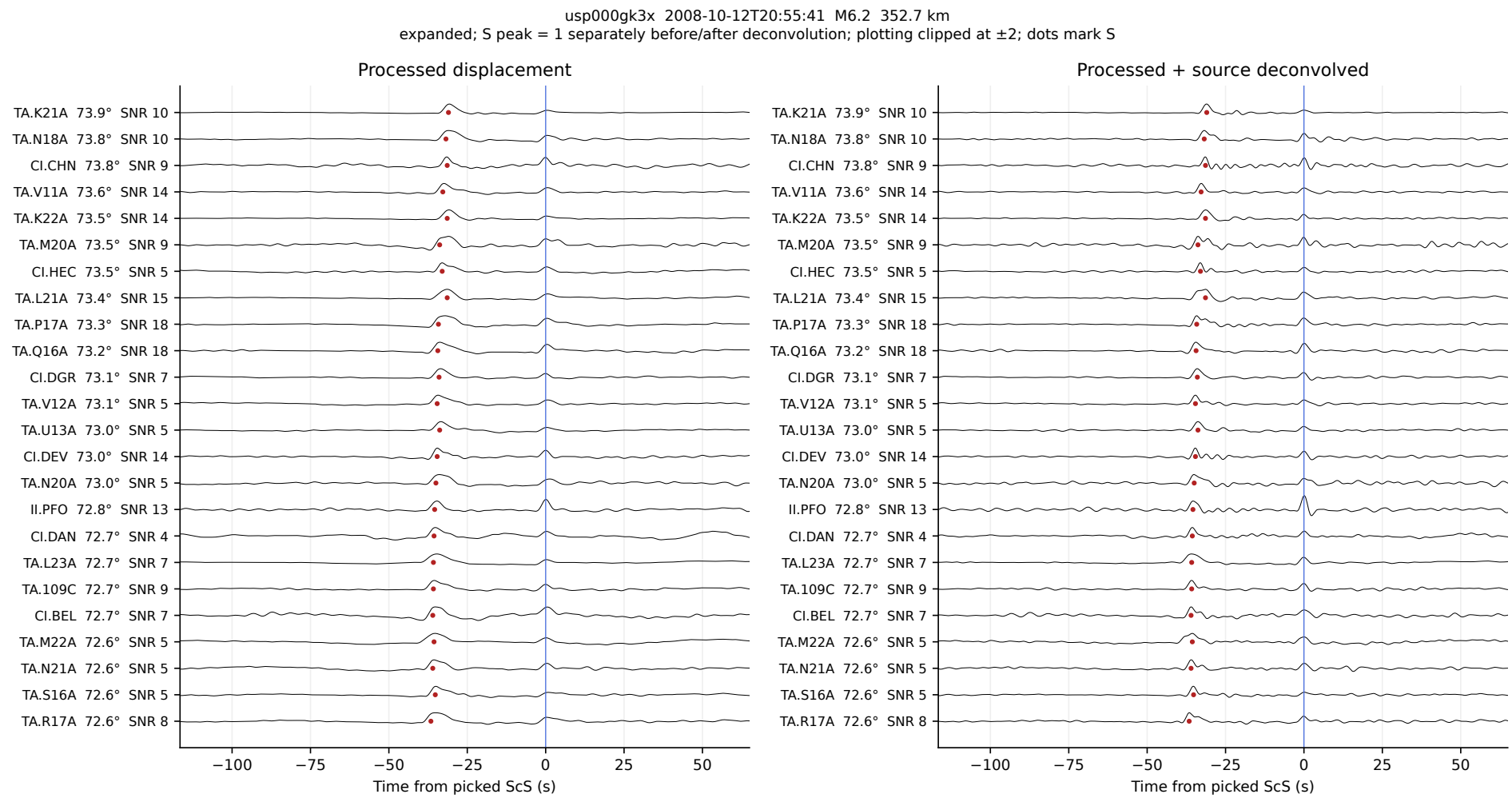
usp000gk3x: 130 used traces; page 1/6

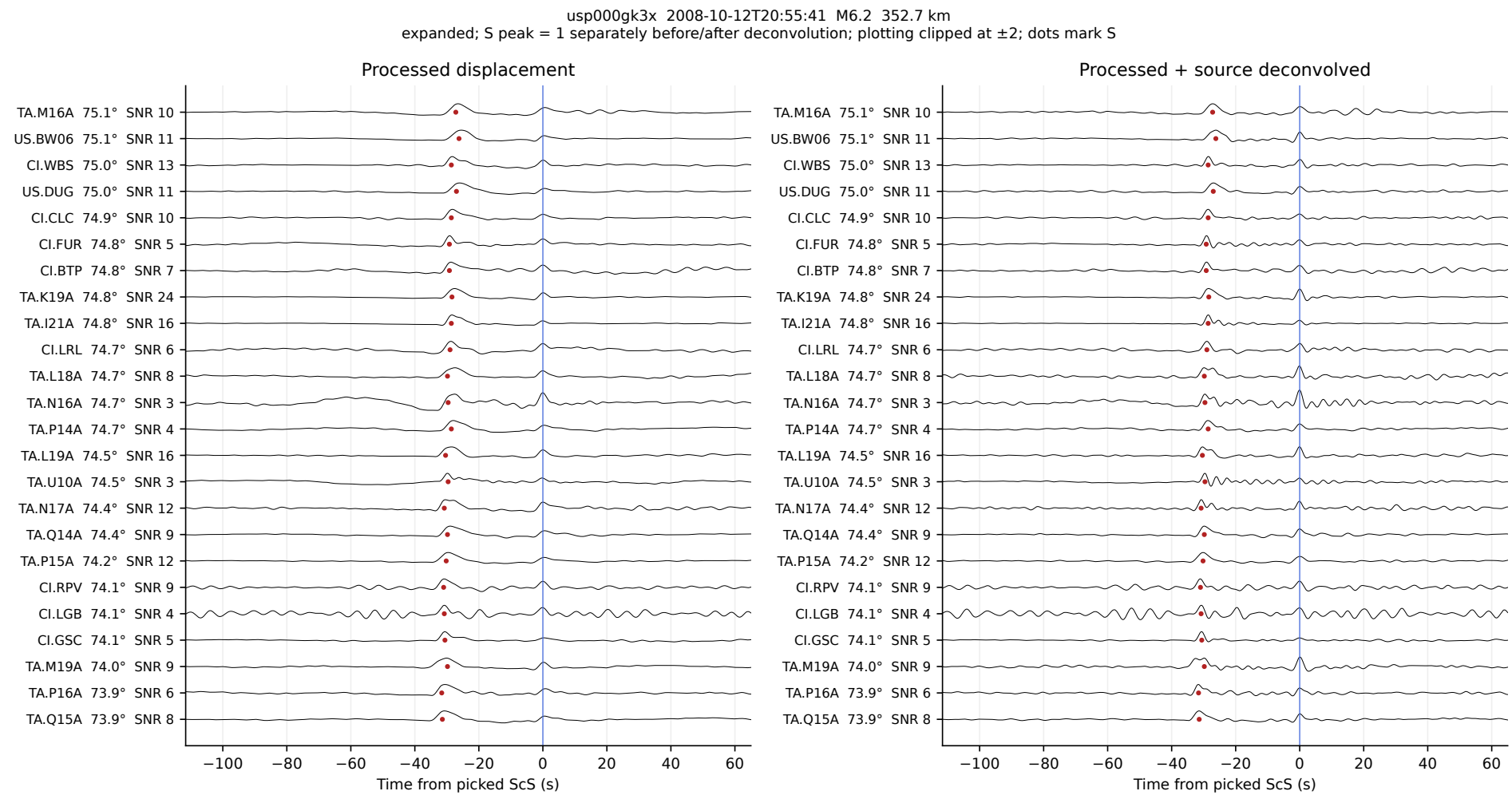


usp000gk3x: 130 used traces; page 2/6

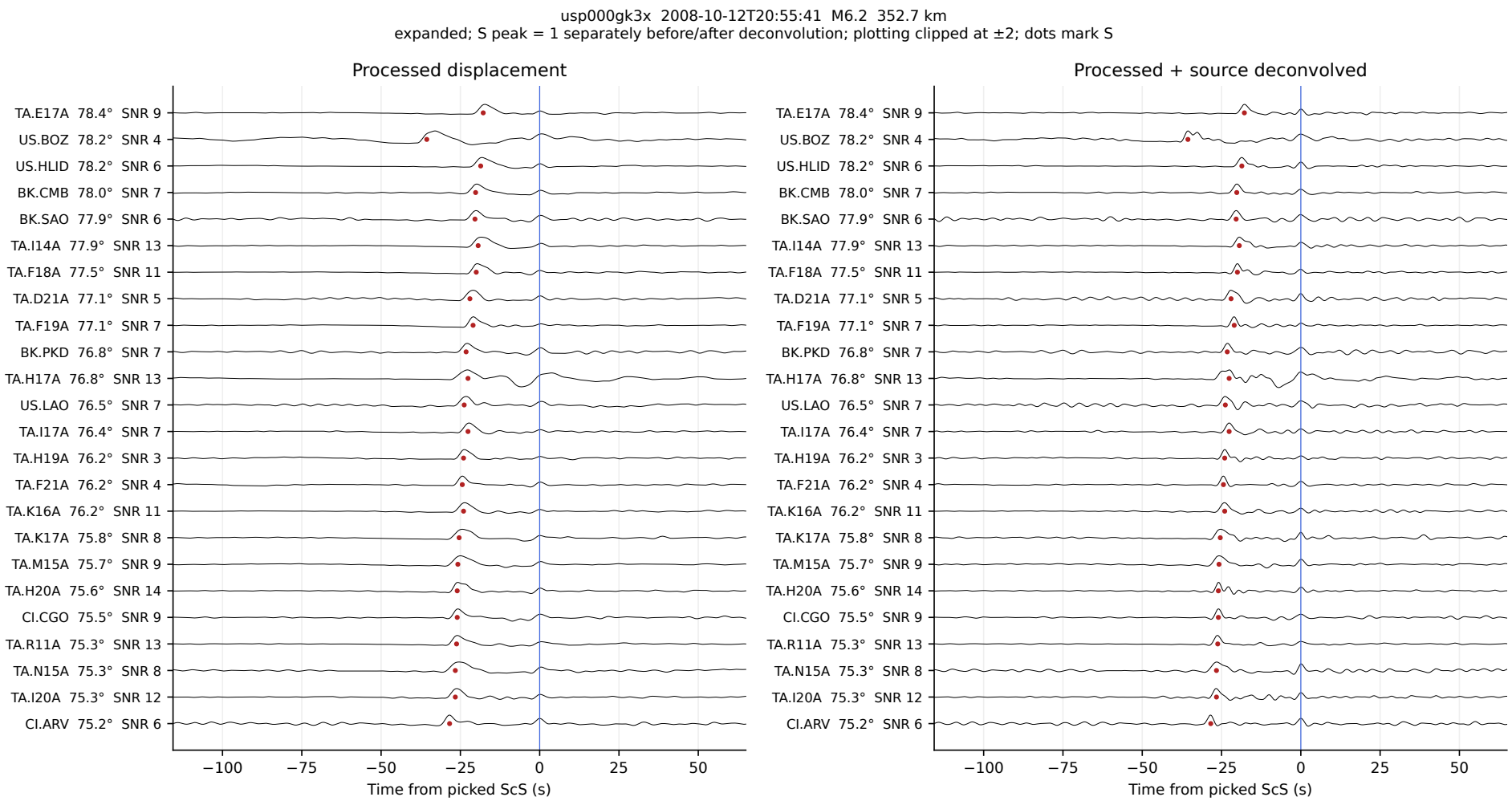


usp000gk3x: 130 used traces; page 3/6

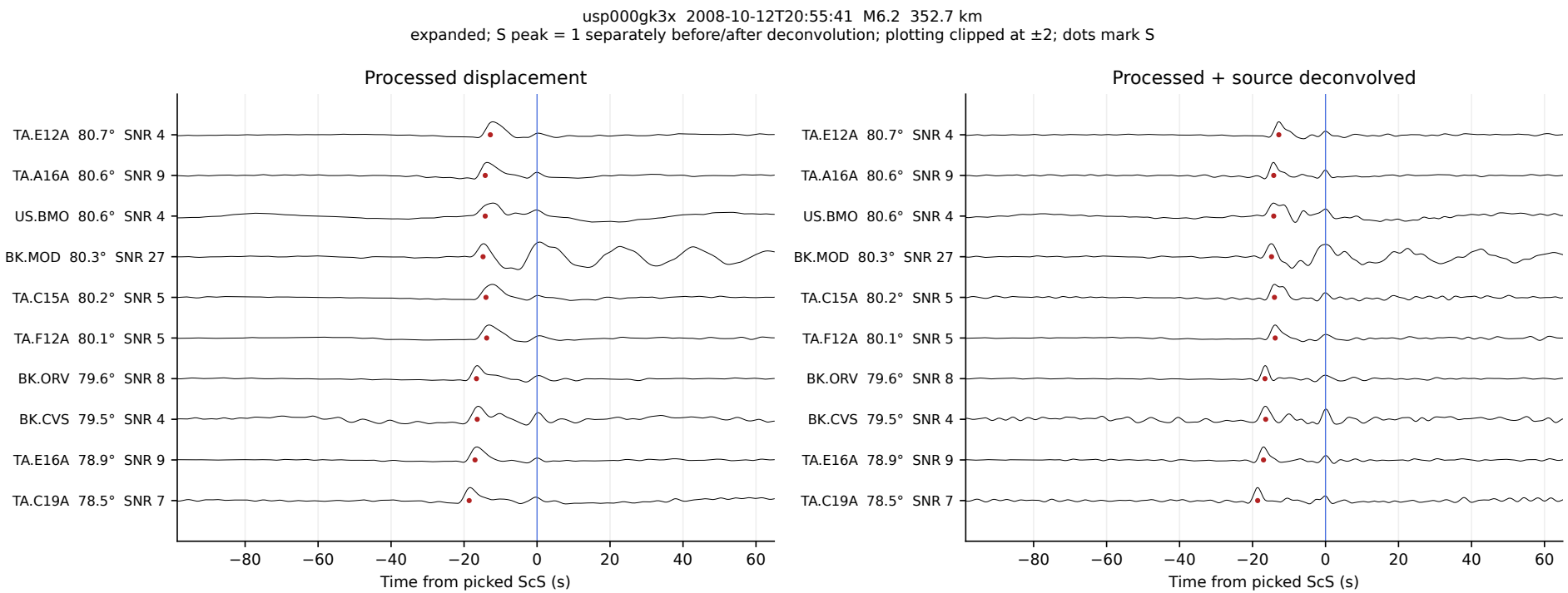



usp000gk3x 2008-10-12T20:55:41 M6.2 352.7 km
expanded; S peak = 1 separately before/after deconvolution; plotting clipped at ±2; dots mark S
Processed displacement
Processed + source deconvolved
TA.M16A 75.1° SNR 10
US.BW06 75.1° SNR 11
CI.WBS 75.0° SNR 13
US.DUG 75.0° SNR 11
CI.CLC 74.9° SNR 10
CI.FUR 74.8° SNR 5
CI.BTP 74.8° SNR 7
TA.K19A 74.8° SNR 24
TA.I21A 74.8° SNR 16
CI.LRL 74.7° SNR 6
TA.L18A 74.7° SNR 8
TA.N16A 74.7° SNR 3
TA.P14A 74.7° SNR 4
TA.L19A 74.5° SNR 16
TA.U10A 74.5° SNR 3
TA.N17A 74.4° SNR 12
TA.Q14A 74.4° SNR 9
TA.P15A 74.2° SNR 12
CI.RPV 74.1° SNR 9
CI.LGB 74.1° SNR 4
CI.GSC 74.1° SNR 5
TA.M19A 74.0° SNR 9
TA.P16A 73.9° SNR 6
TA.Q15A 73.9° SNR 8
−100 −80 −60 −40 −20 0 20 40 60
Time from picked ScS (s)

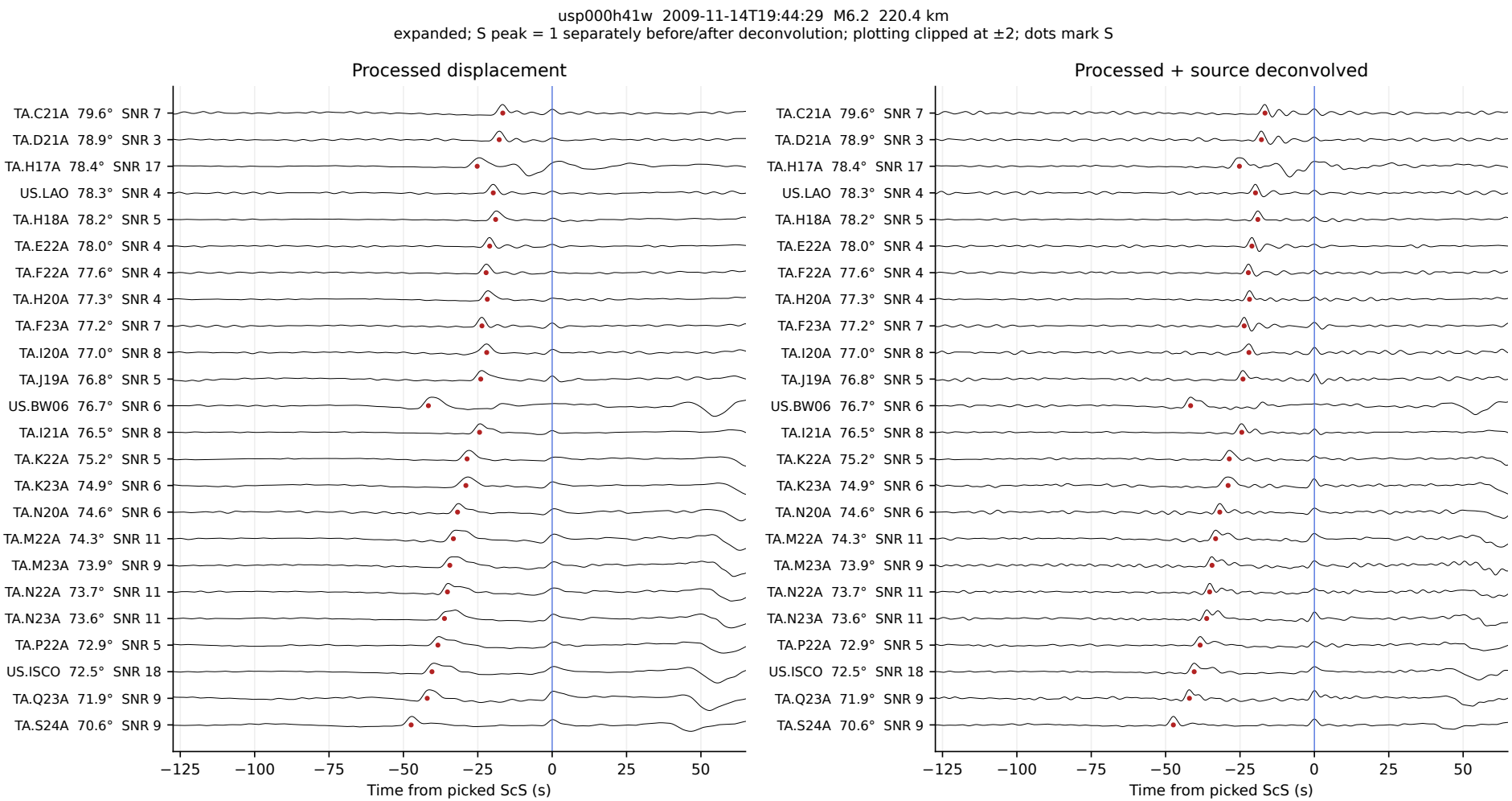
usp000gk3x: 130 used traces; page 5/6



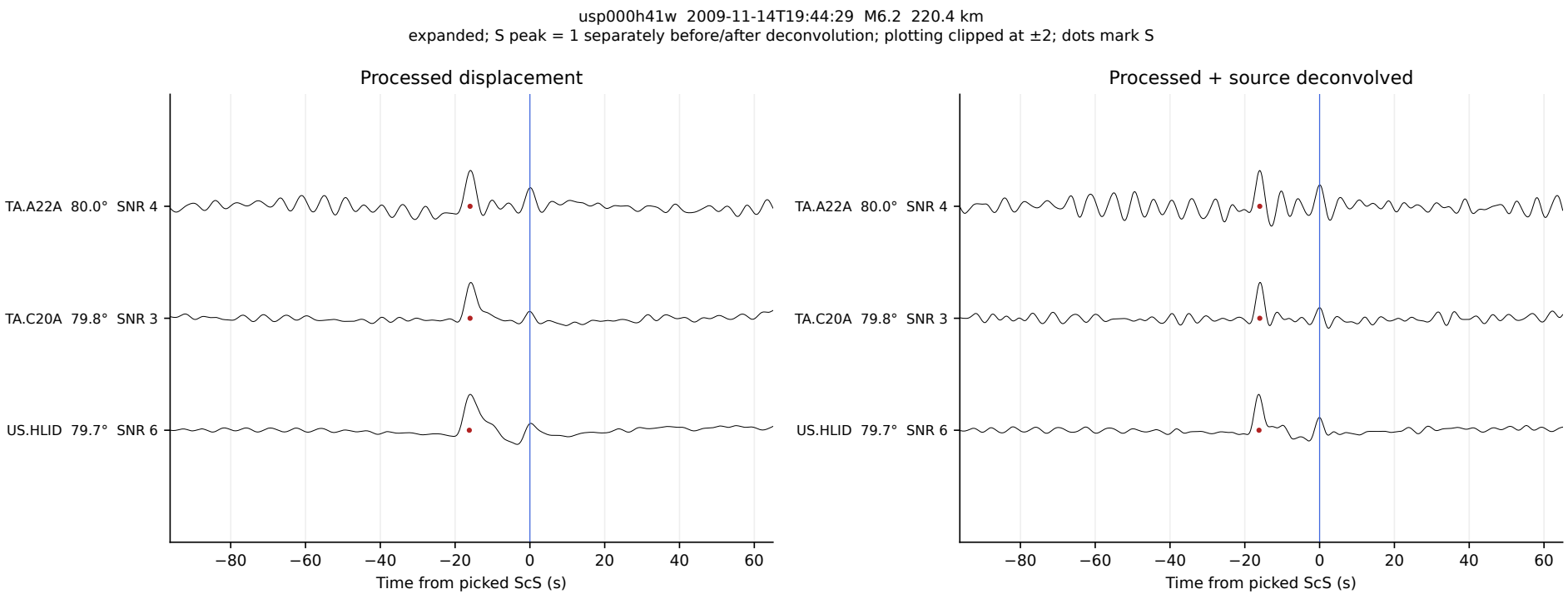
usp000gk3x: 130 used traces; page 6/6



usp000h41w: 27 used traces; page 1/2

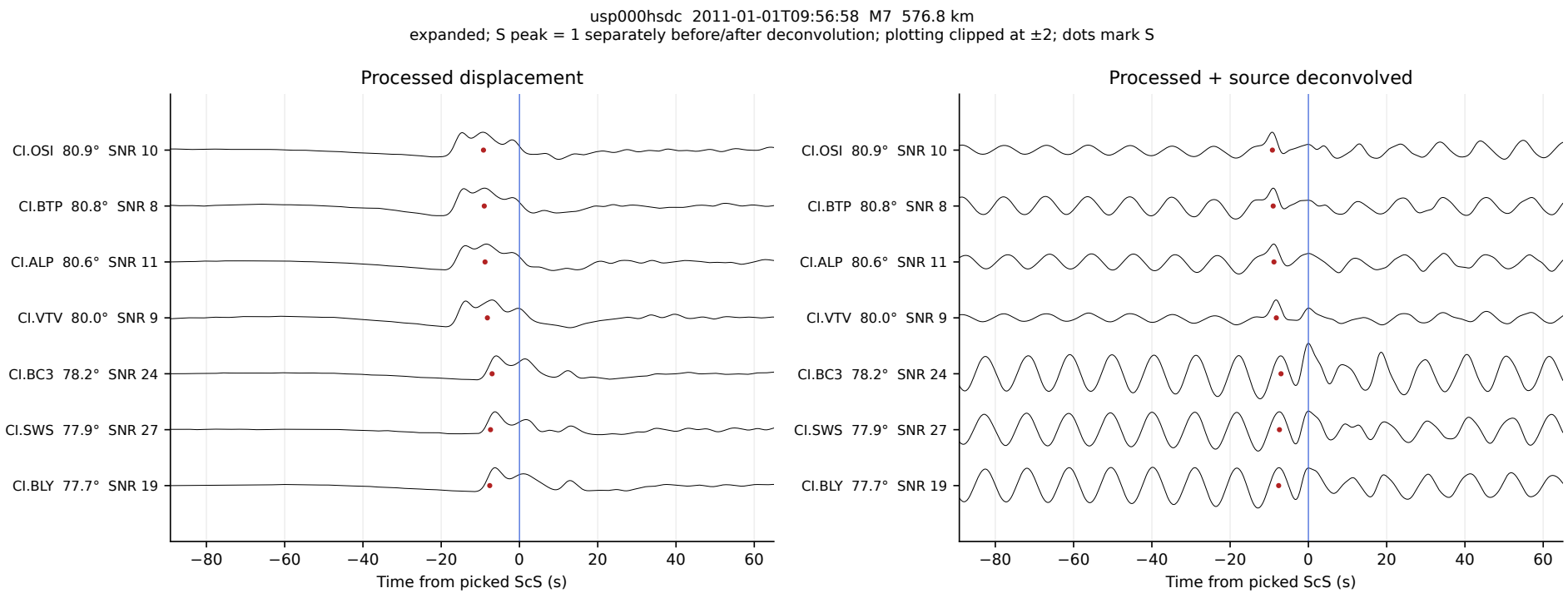


usp000h41w: 27 used traces; page 2/2



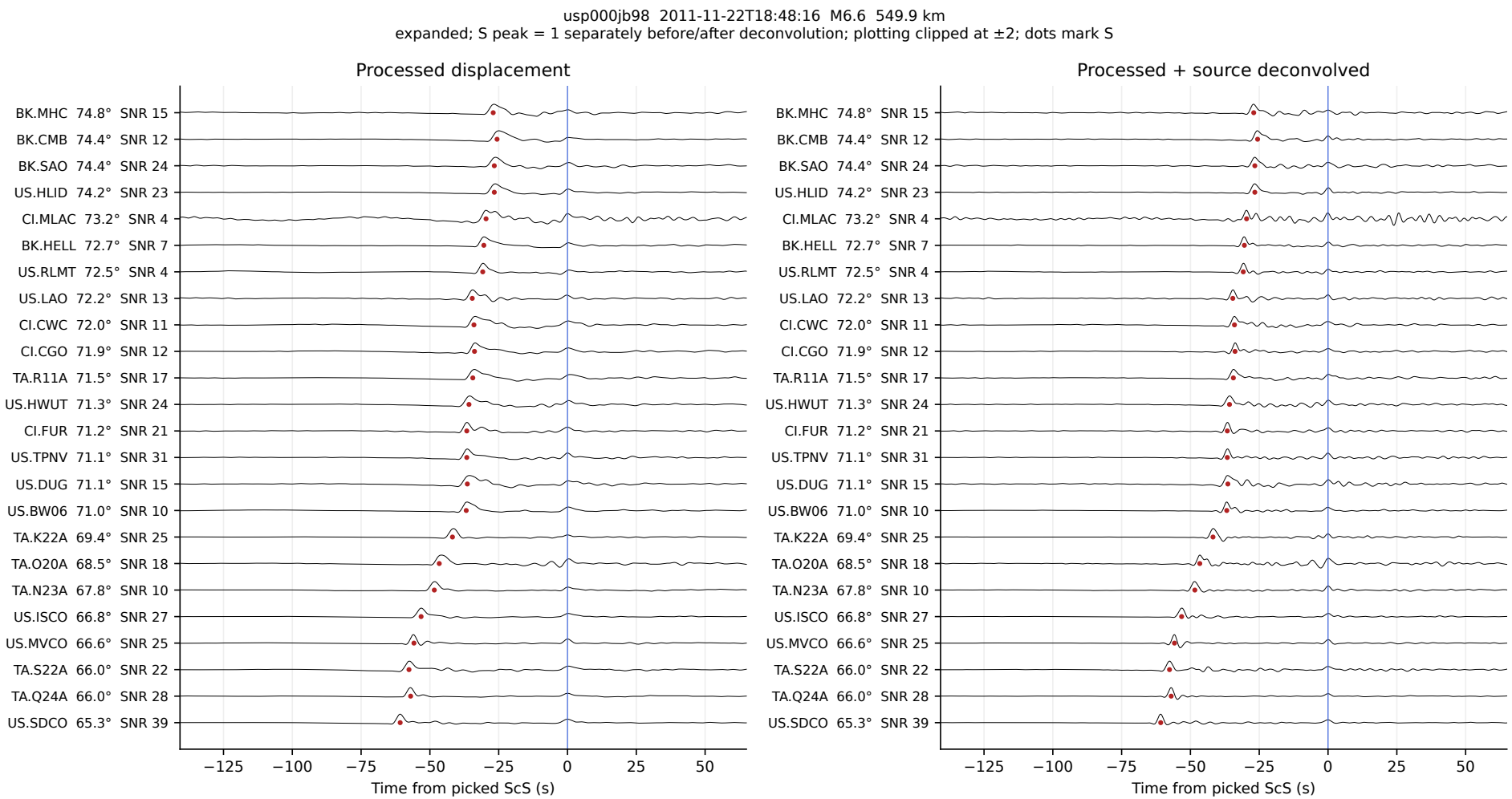
usp000hd4k / 2010-05-24T16:18:29.090000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 37

usp000hsdc: 7 used traces; page 1/1

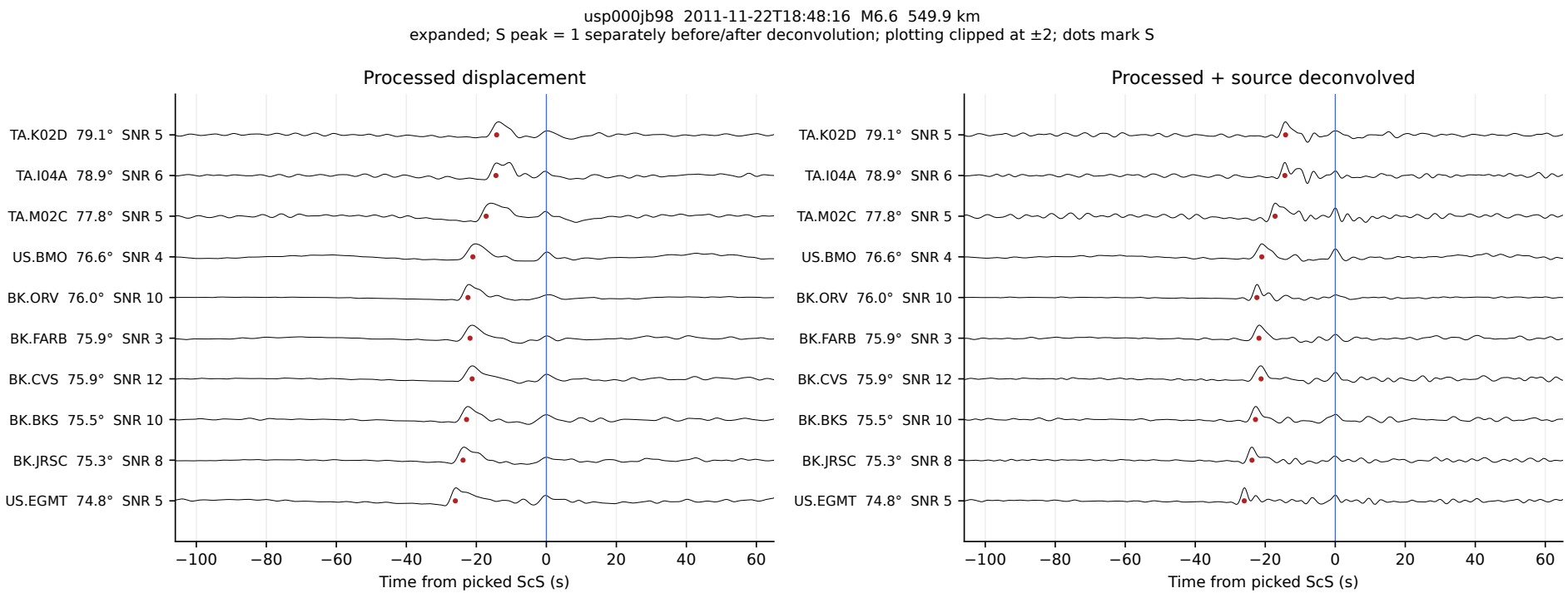


usp000j7ee / 2011-09-02T13:47:09.620000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 89

usp000jb98: 34 used traces; page 1/2

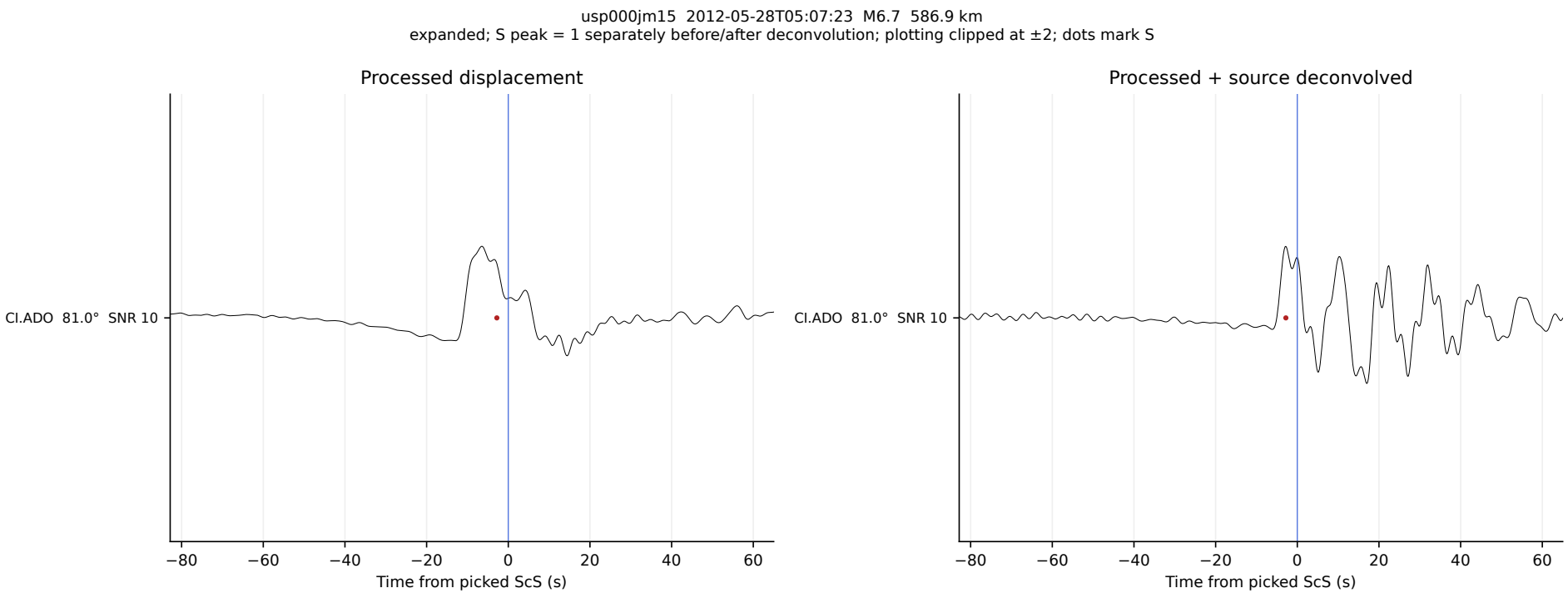


usp000jb98: 34 used traces; page 2/2

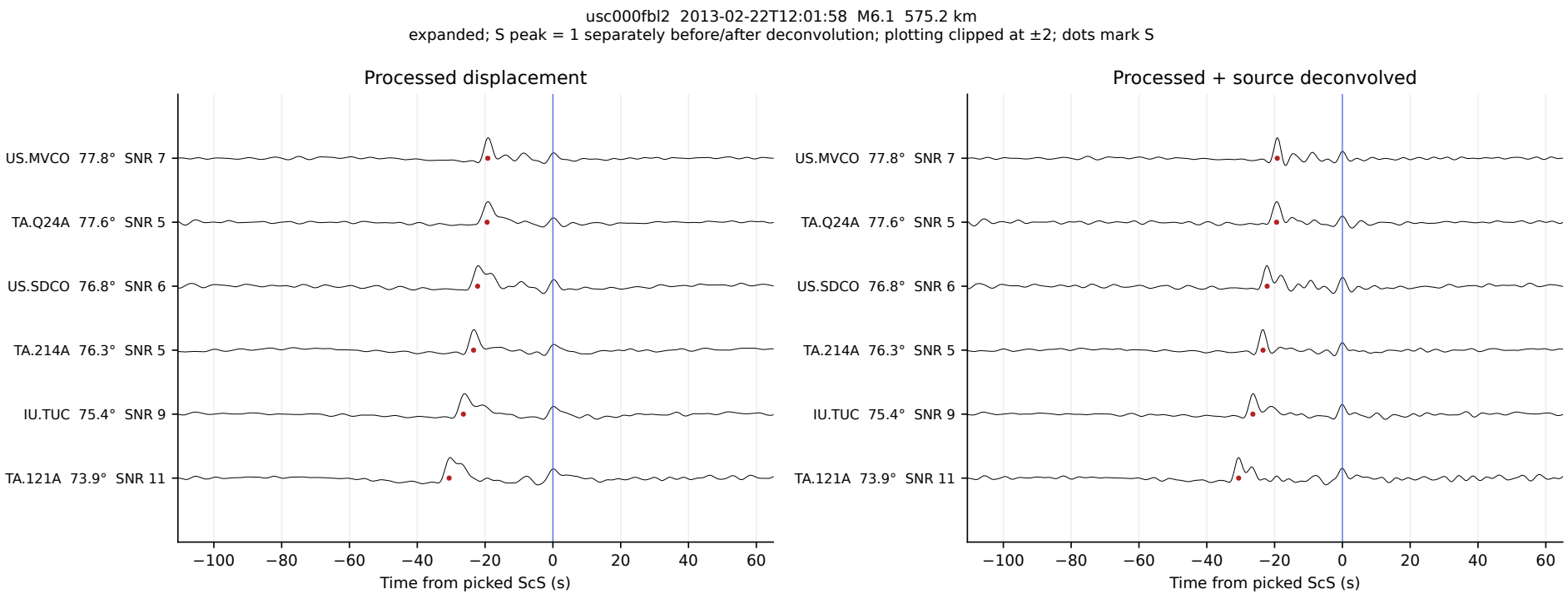


usp000jftj / 2012-03-05T07:46:10.040000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 90

usp000jm15: 1 used traces; page 1/1



usc000fbl2: 6 used traces; page 1/1

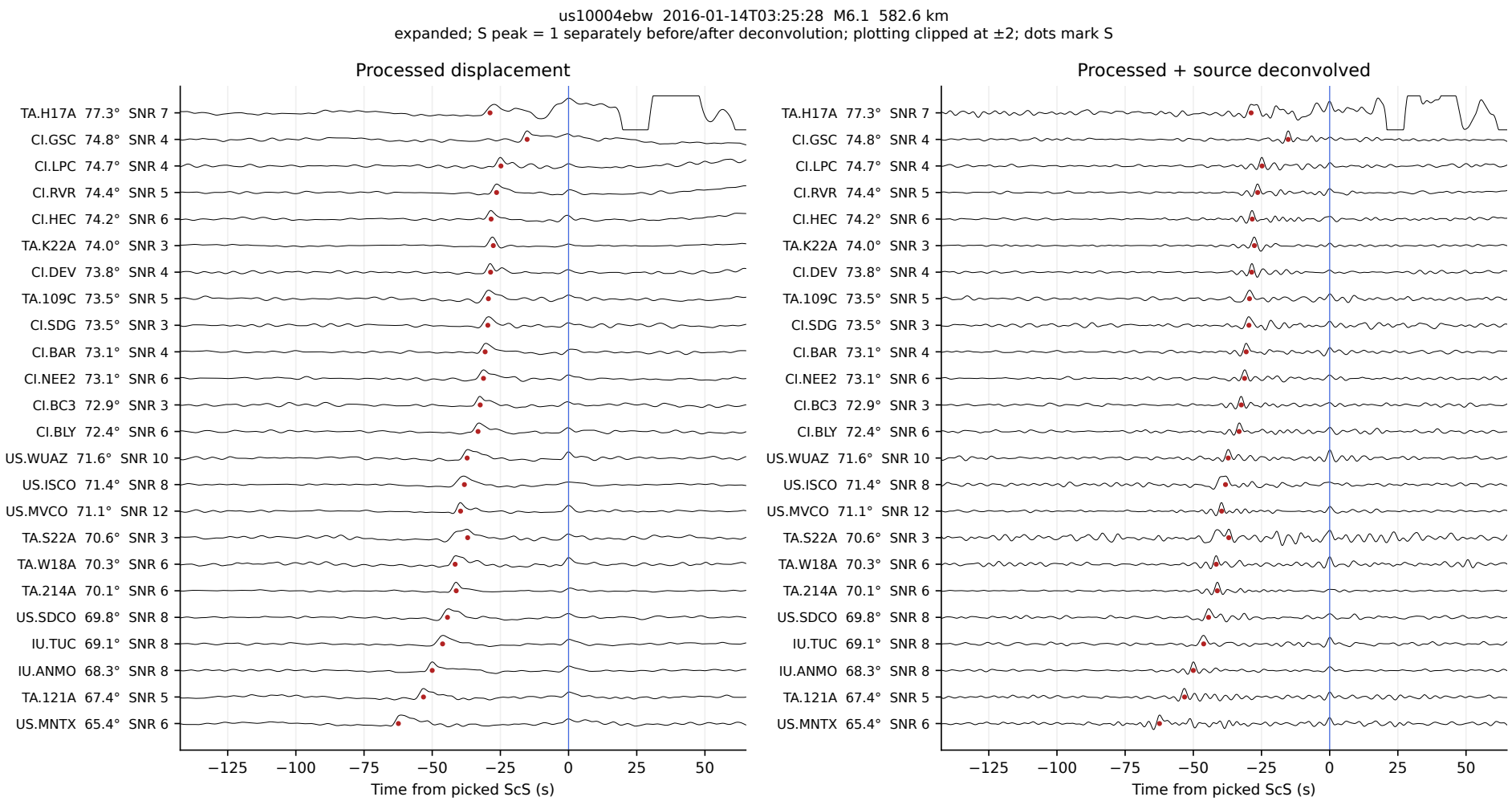


usb000sfj5 / 2014-09-24T11:16:15.010000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 118

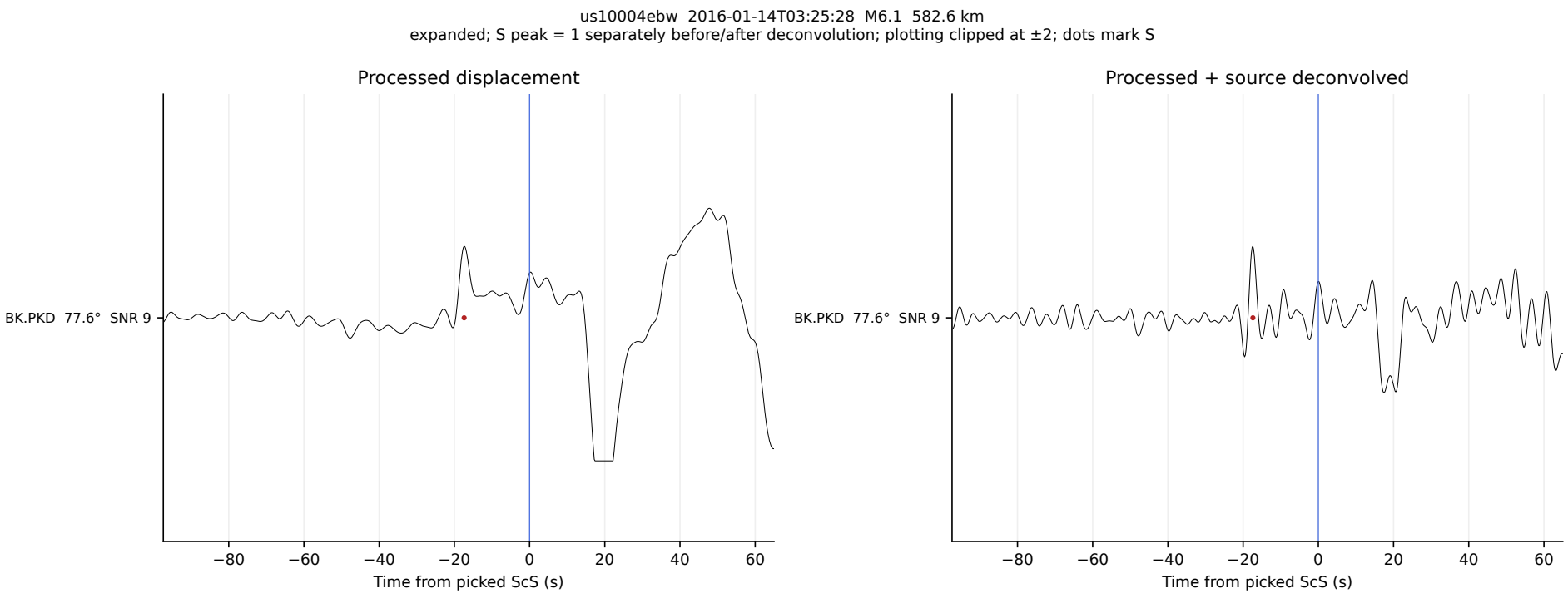
usc000tp1s / 2015-02-11T18:57:22.460000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 123

us100041f1 / 2015-11-26T05:45:18.400000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 48

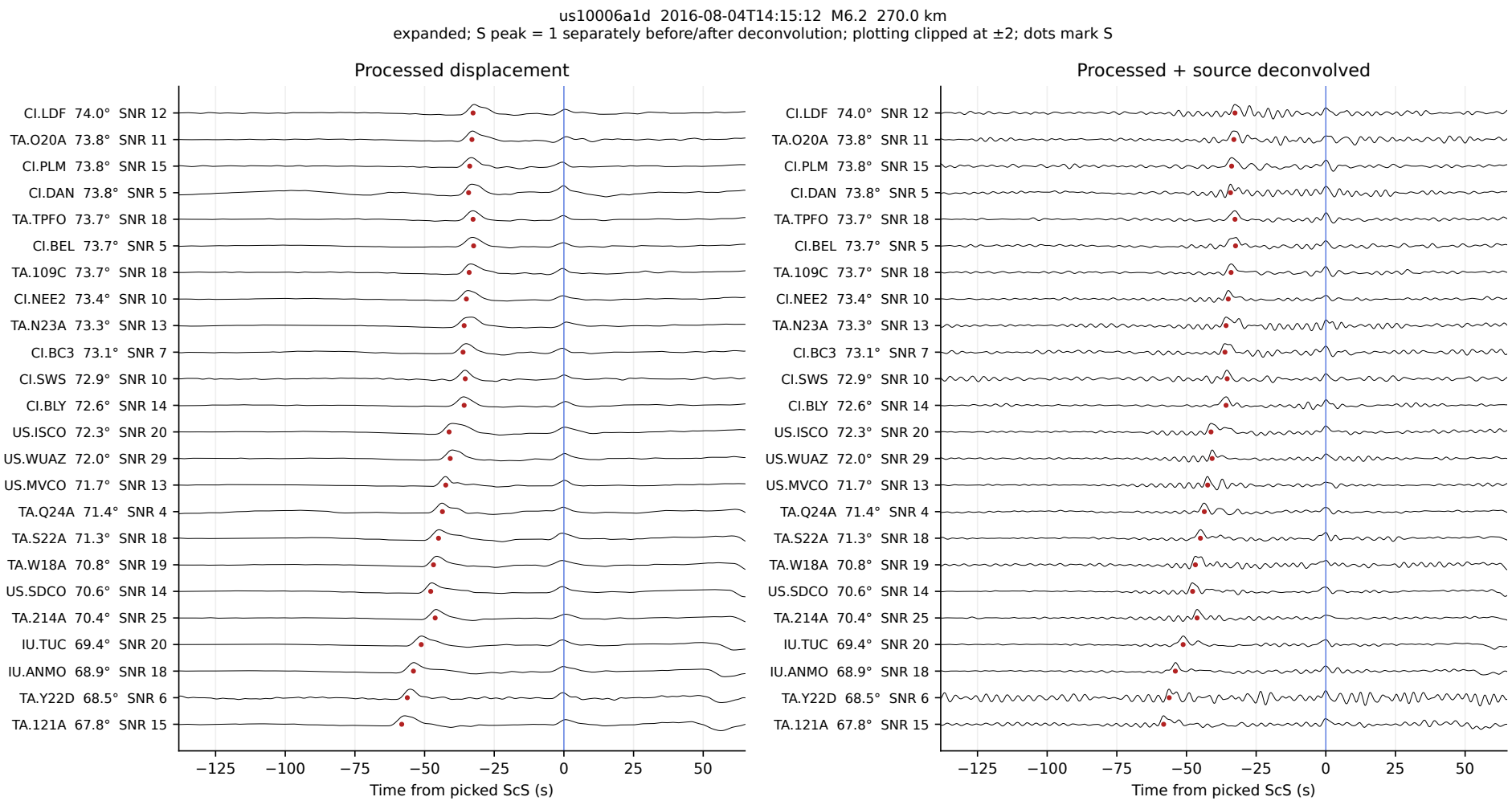
us10004ebw: 25 used traces; page 1/2



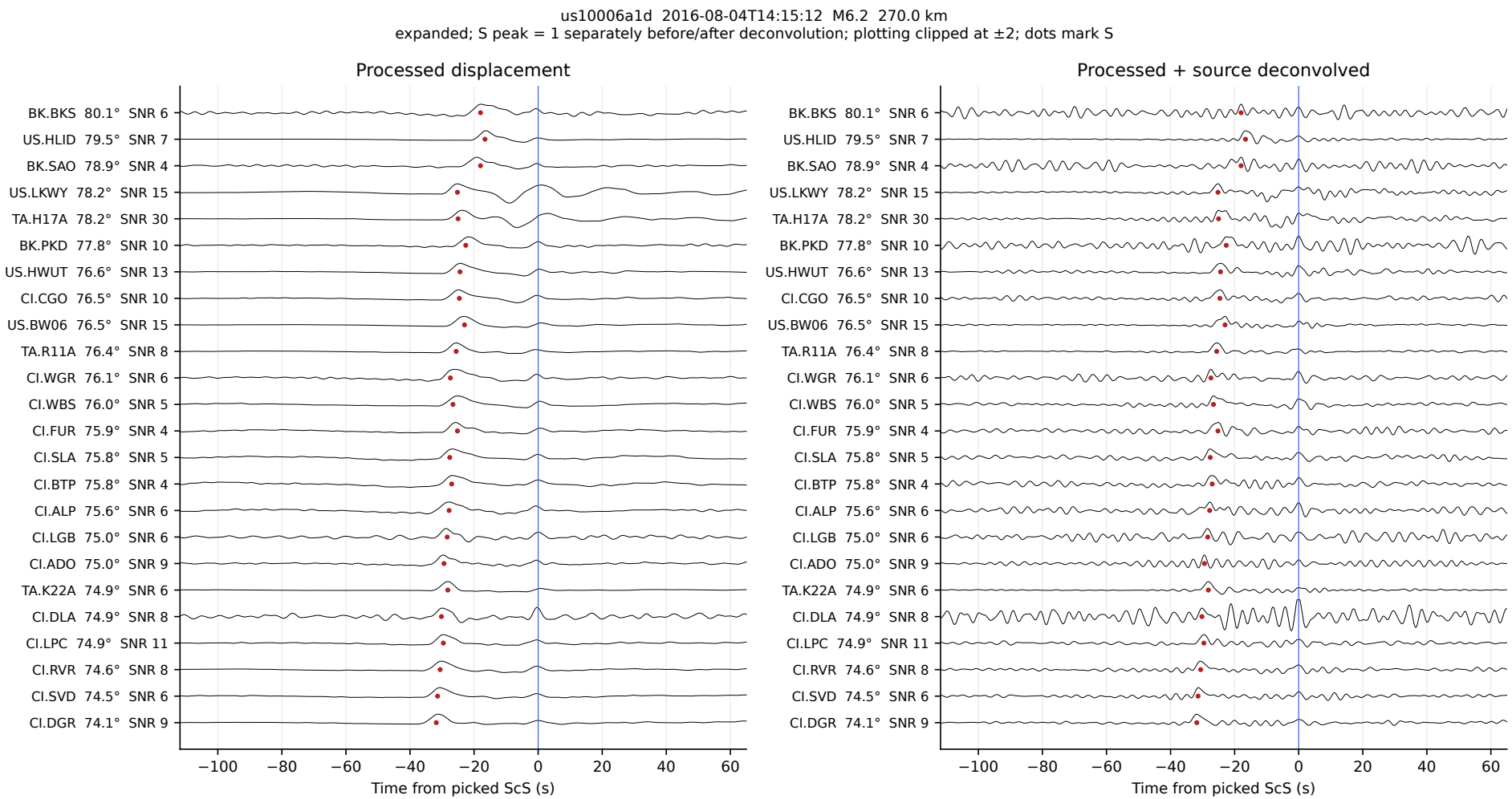
us10004ebw: 25 used traces; page 2/2



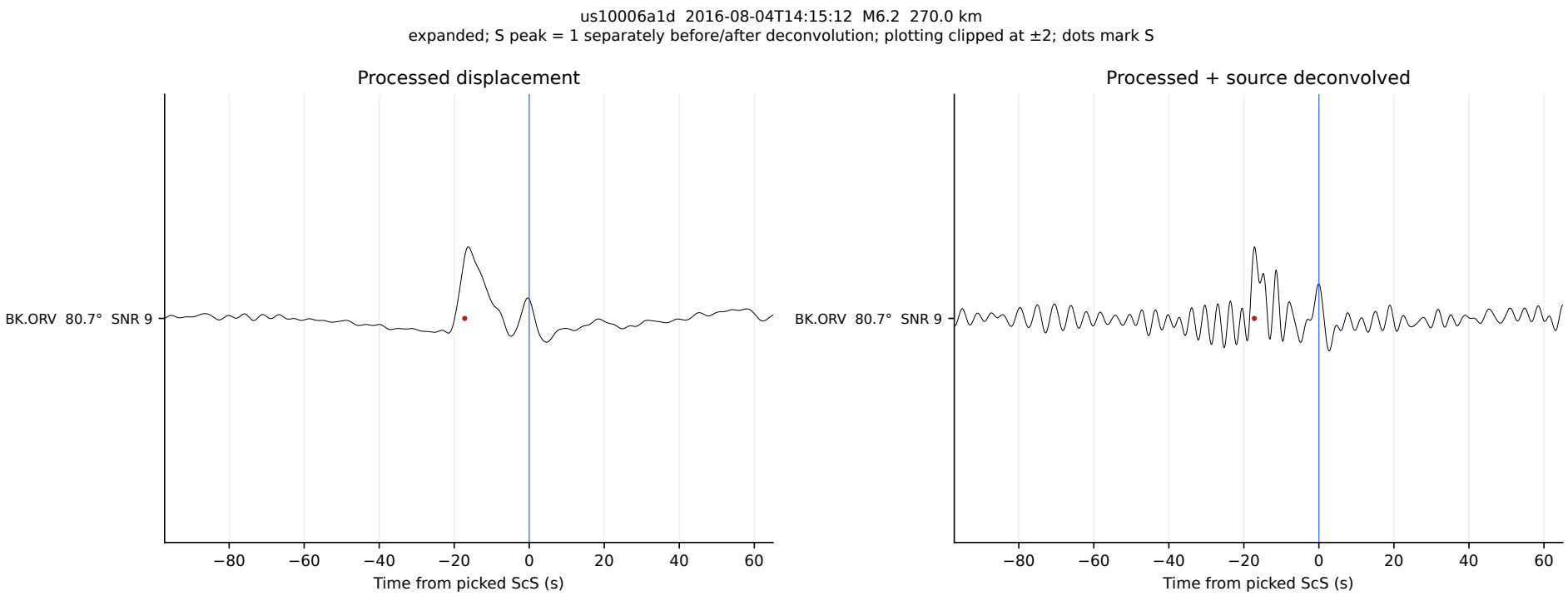
us10006a1d: 49 used traces; page 1/3



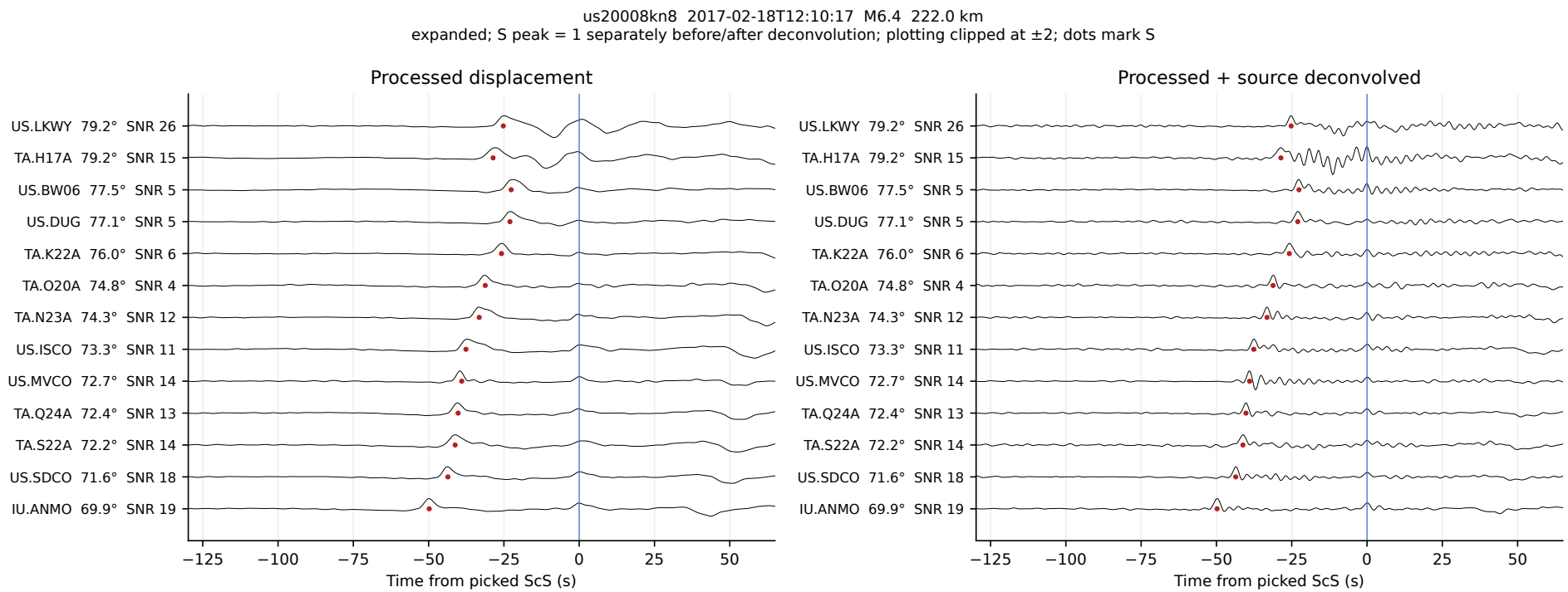
us10006a1d: 49 used traces; page 2/3



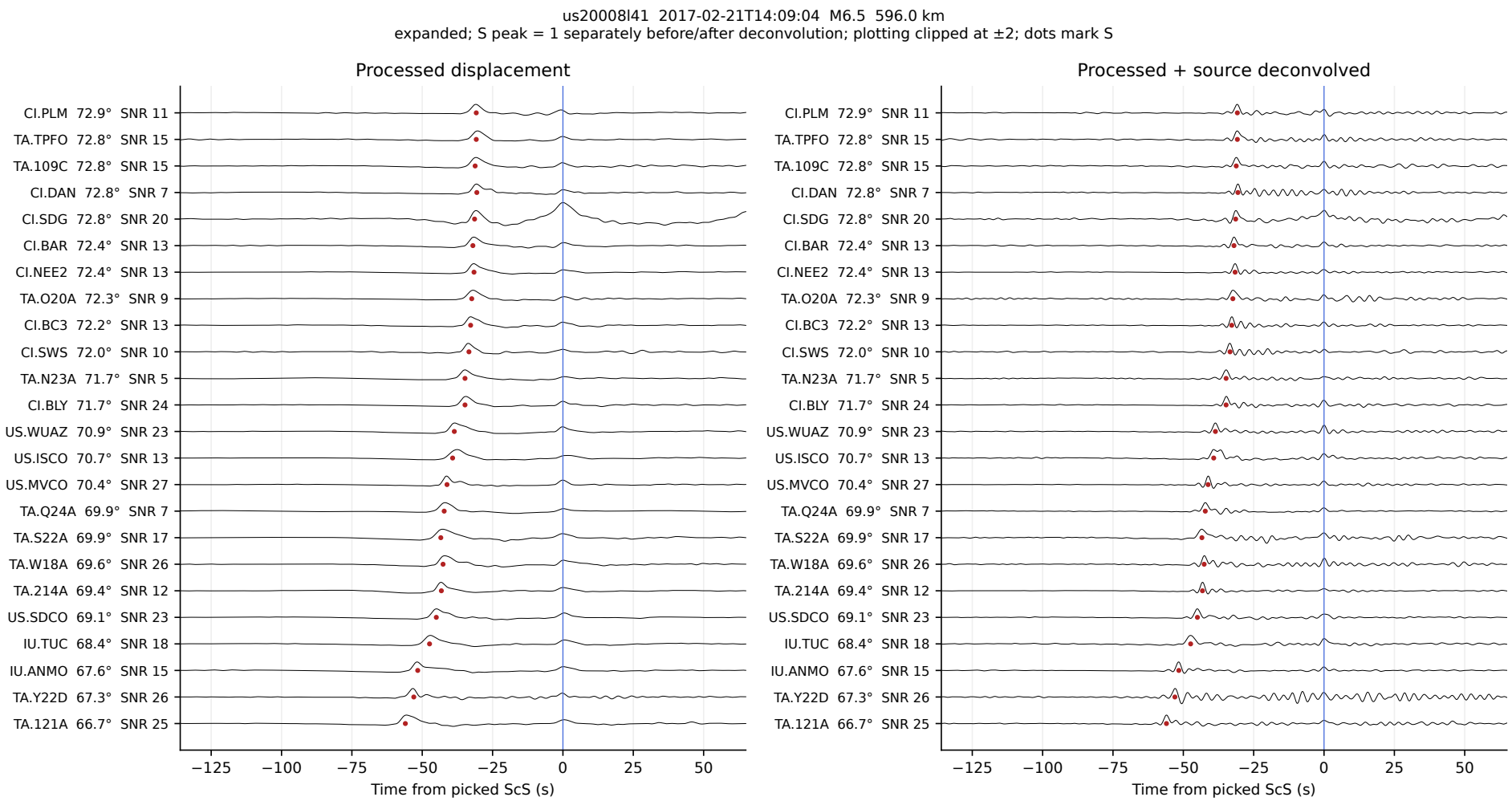
us10006a1d: 49 used traces; page 3/3

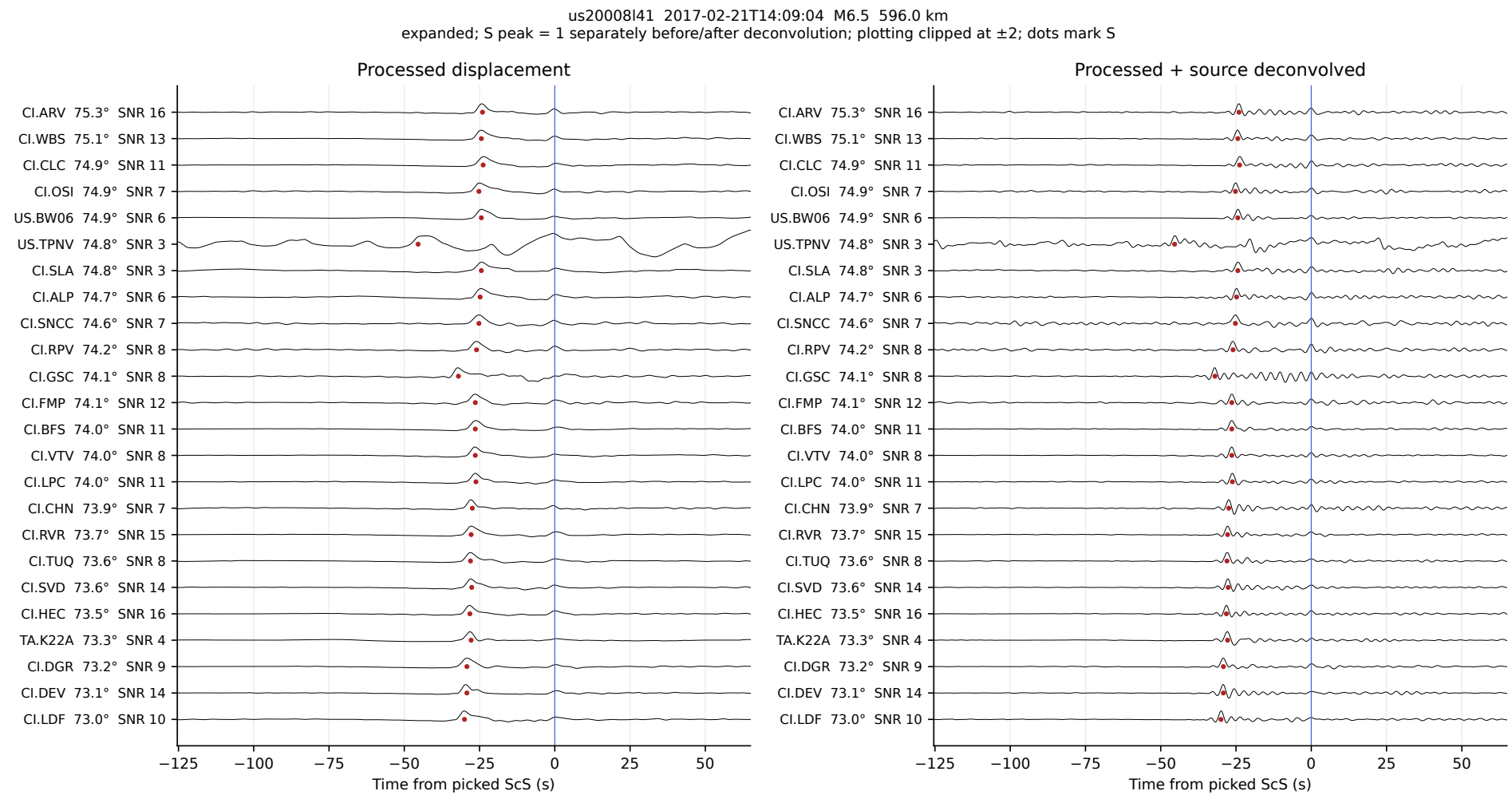


us20008kn8: 13 used traces; page 1/1

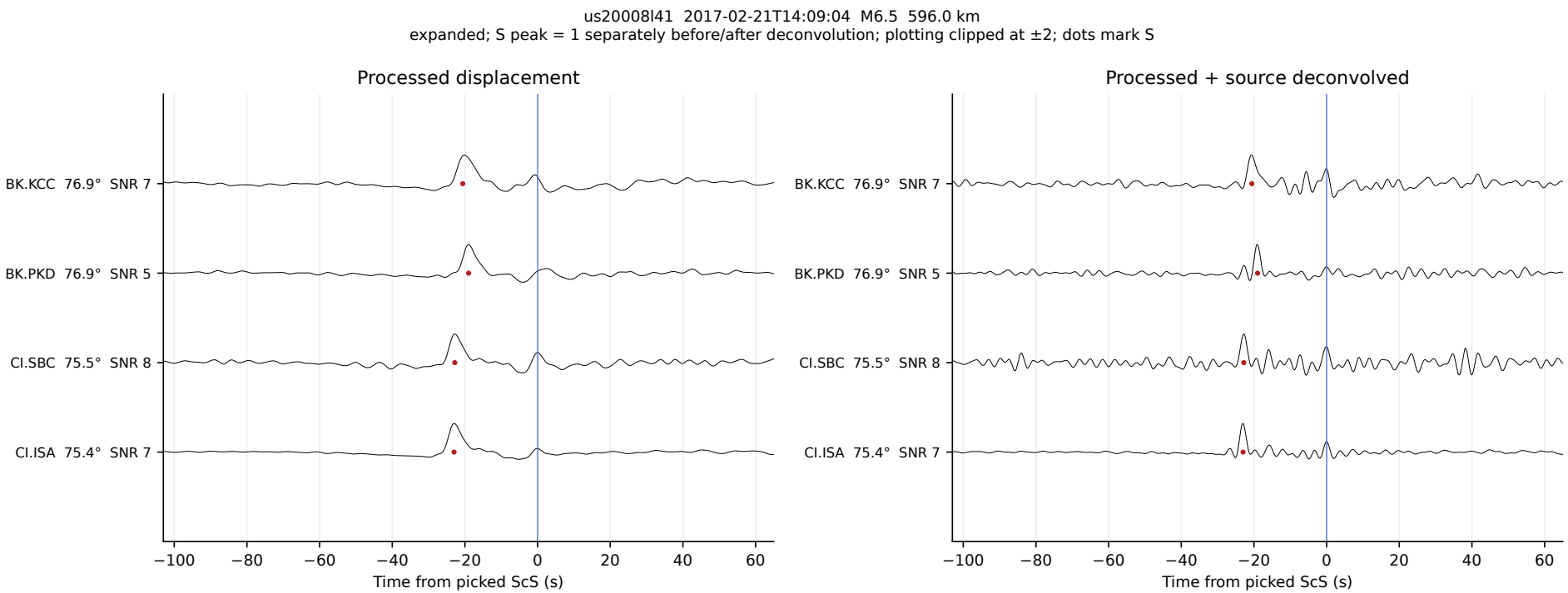


us20008l41: 52 used traces; page 1/3




us20008l41 2017-02-21T14:09:04 M6.5 596.0 km
expanded; S peak = 1 separately before/after deconvolution; plotting clipped at ±2; dots mark S
Processed displacement
Processed + source deconvolved
CI.ARV 75.3° SNR 16
CI.WBS 75.1° SNR 13
CI.CLC 74.9° SNR 11
CI.OSI 74.9° SNR 7
US.BW06 74.9° SNR 6
US.TPNV 74.8° SNR 3
CI.SLA 74.8° SNR 3
CI.ALP 74.7° SNR 6
CI.SNCC 74.6° SNR 7
CI.RPV 74.2° SNR 8
CI.GSC 74.1° SNR 8
CI.FMP 74.1° SNR 12
CI.BFS 74.0° SNR 11
CI.VTV 74.0° SNR 8
CI.LPC 74.0° SNR 11
CI.CHN 73.9° SNR 7
CI.RVR 73.7° SNR 15
CI.TUQ 73.6° SNR 8
CI.SVD 73.6° SNR 14
CI.HEC 73.5° SNR 16
TA.K22A 73.3° SNR 4
CI.DGR 73.2° SNR 9
CI.DEV 73.1° SNR 14
CI.LDF 73.0° SNR 10
−125 −100 −75 −50 −25 0 25 50
Time from picked ScS (s)

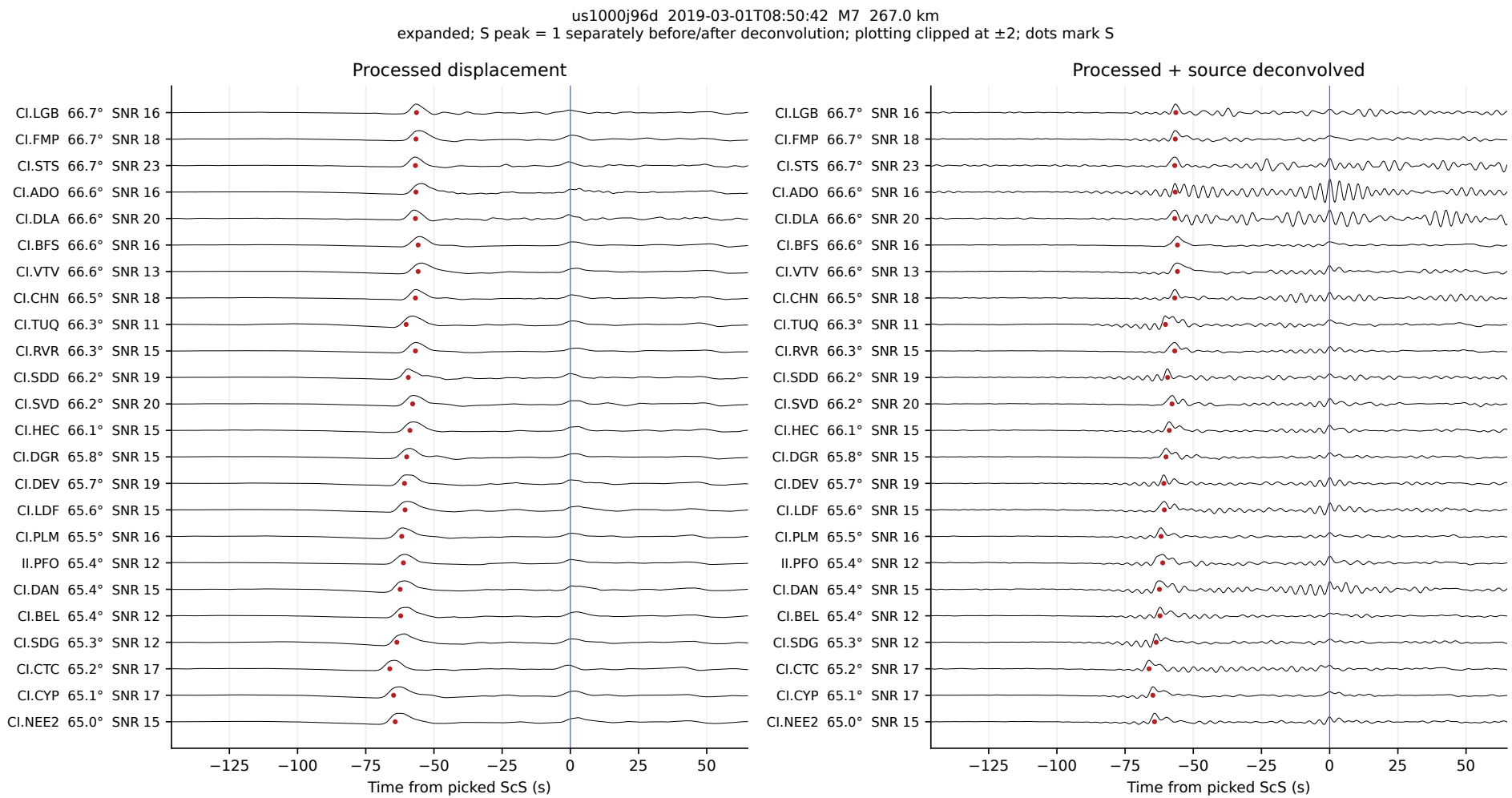
us20008l41: 52 used traces; page 3/3



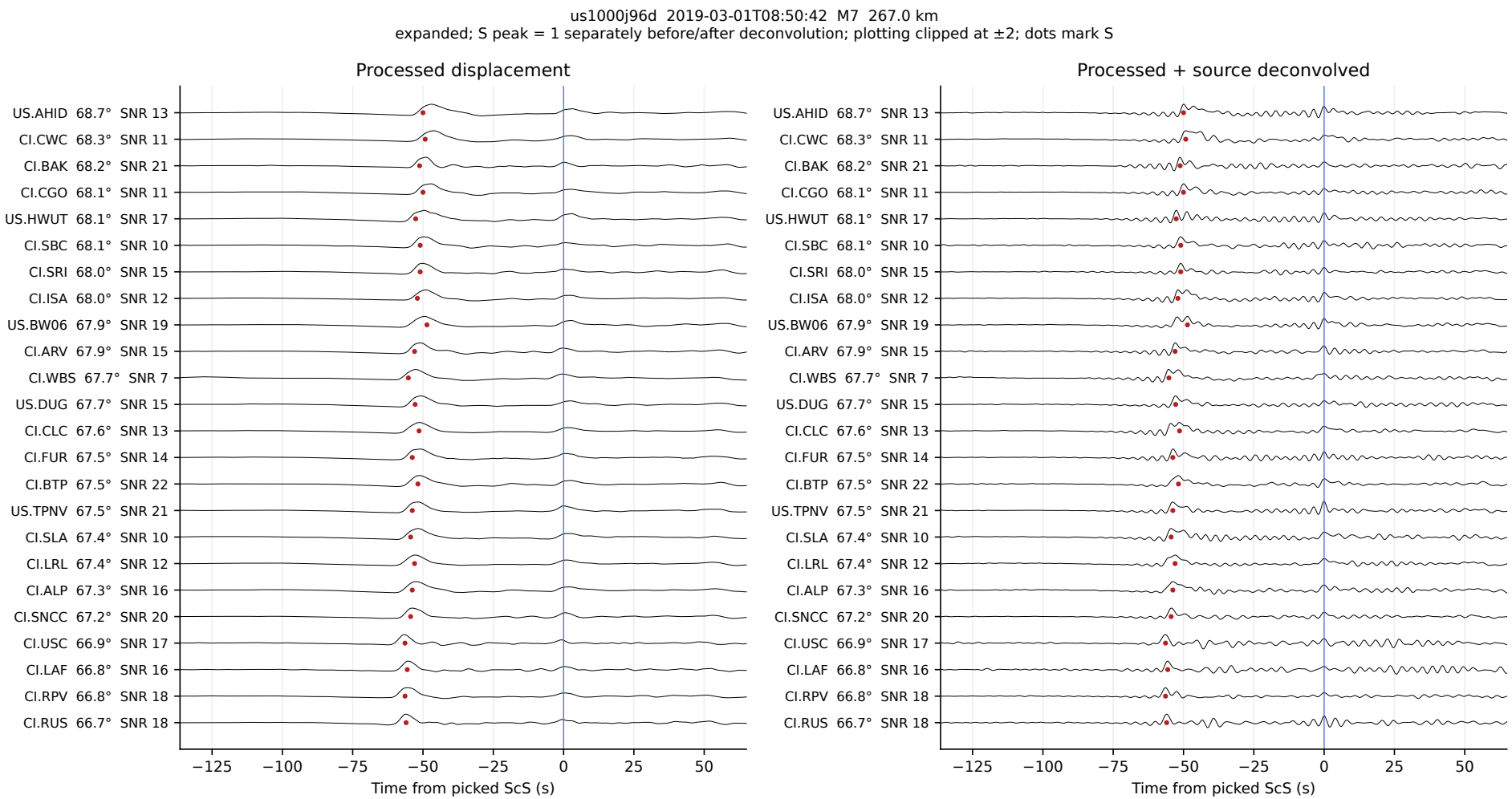
us1000dce2 / 2018-04-02T13:40:34.840000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 108

us2000j0ja / 2019-01-05T19:25:38.520000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 24

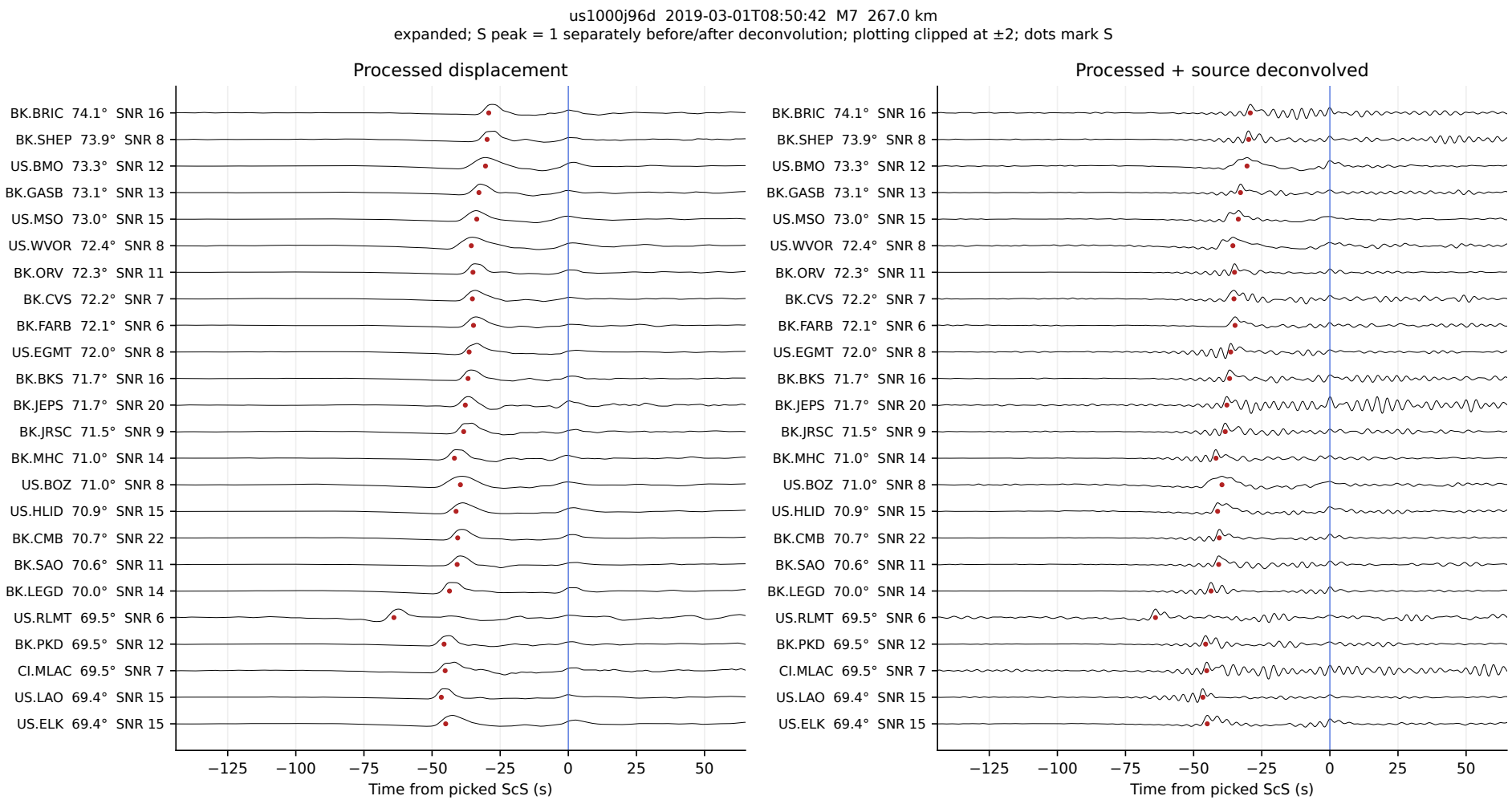
us1000j96d: 75 used traces; page 1/4



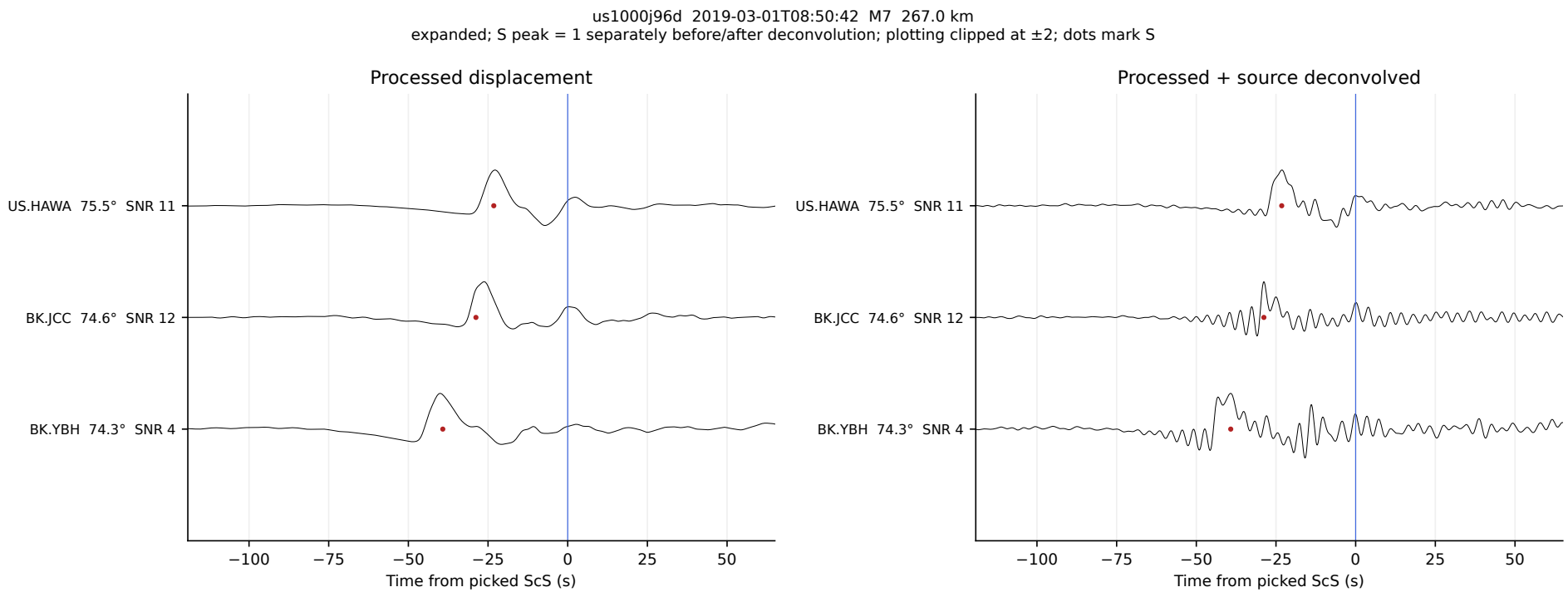
us1000j96d: 75 used traces; page 2/4



us1000j96d: 75 used traces; page 3/4

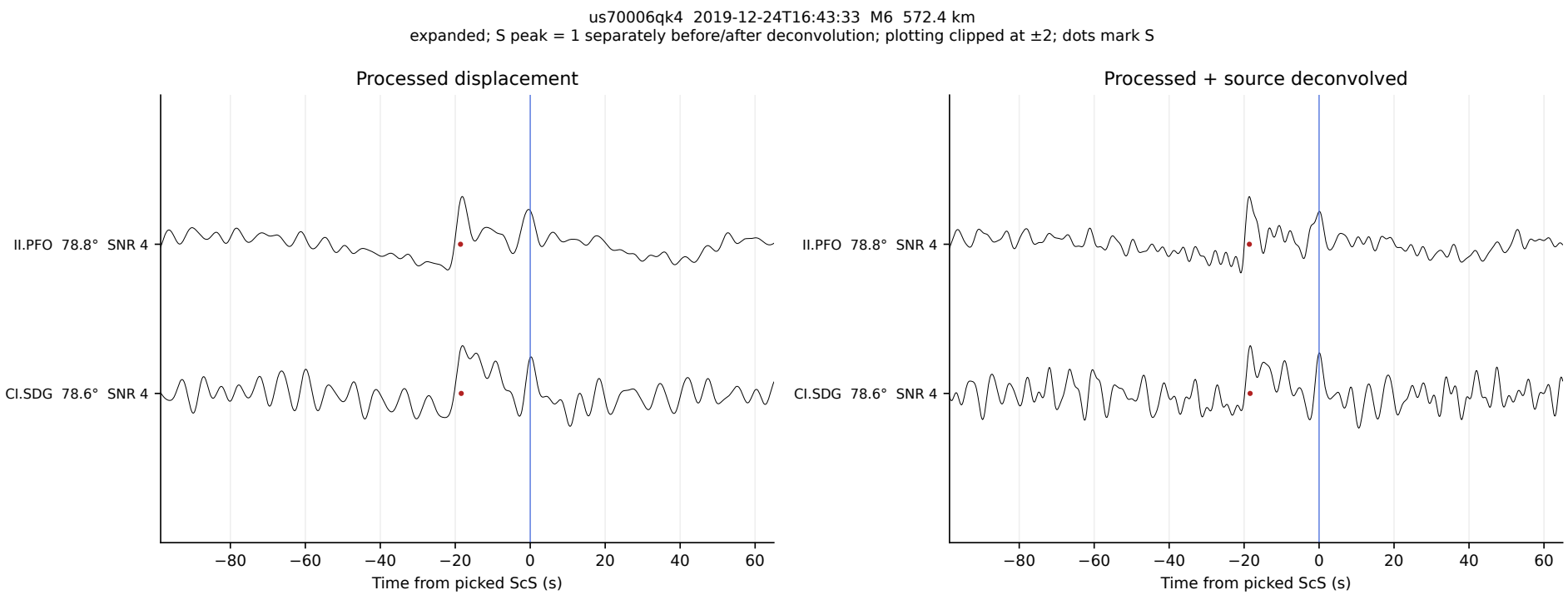


us1000j96d: 75 used traces; page 4/4

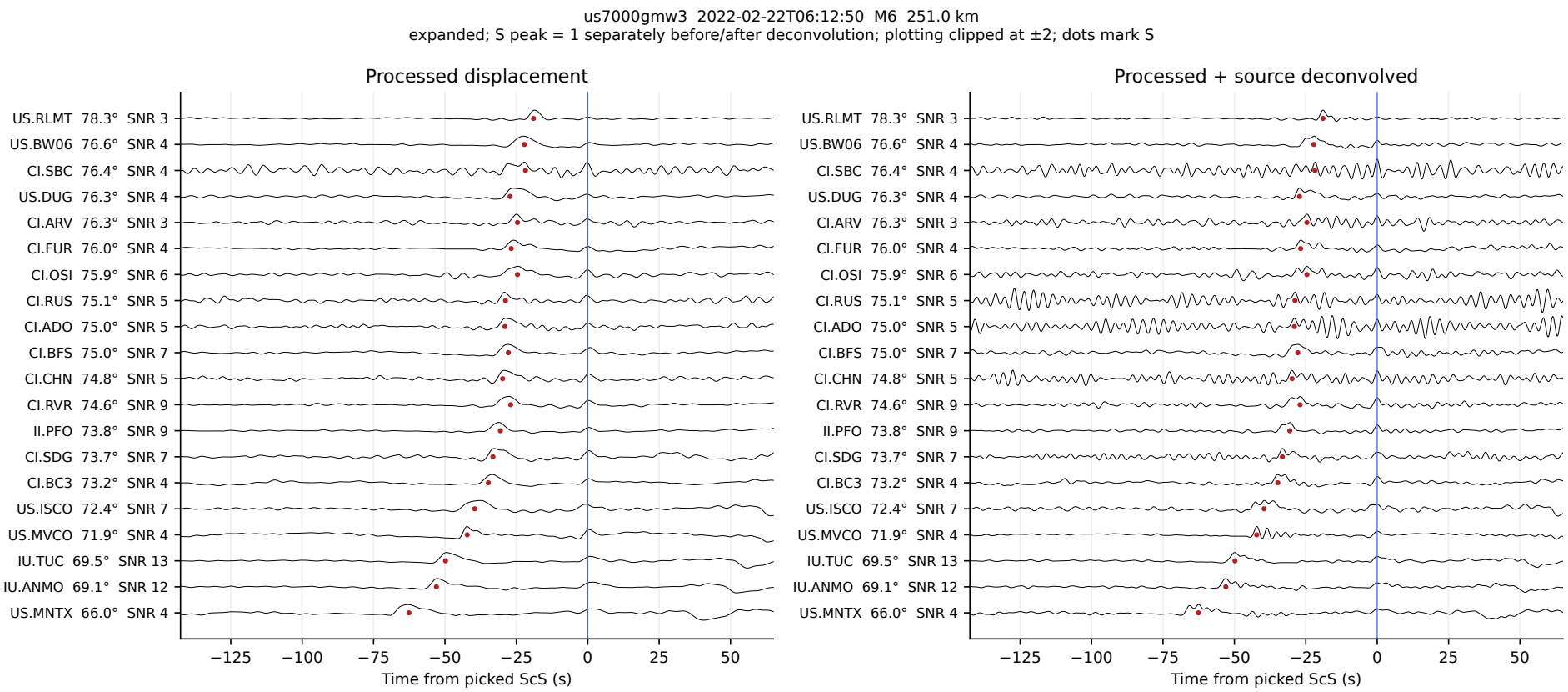


us1000jg5z / 2019-03-15T05:03:50.060000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 98

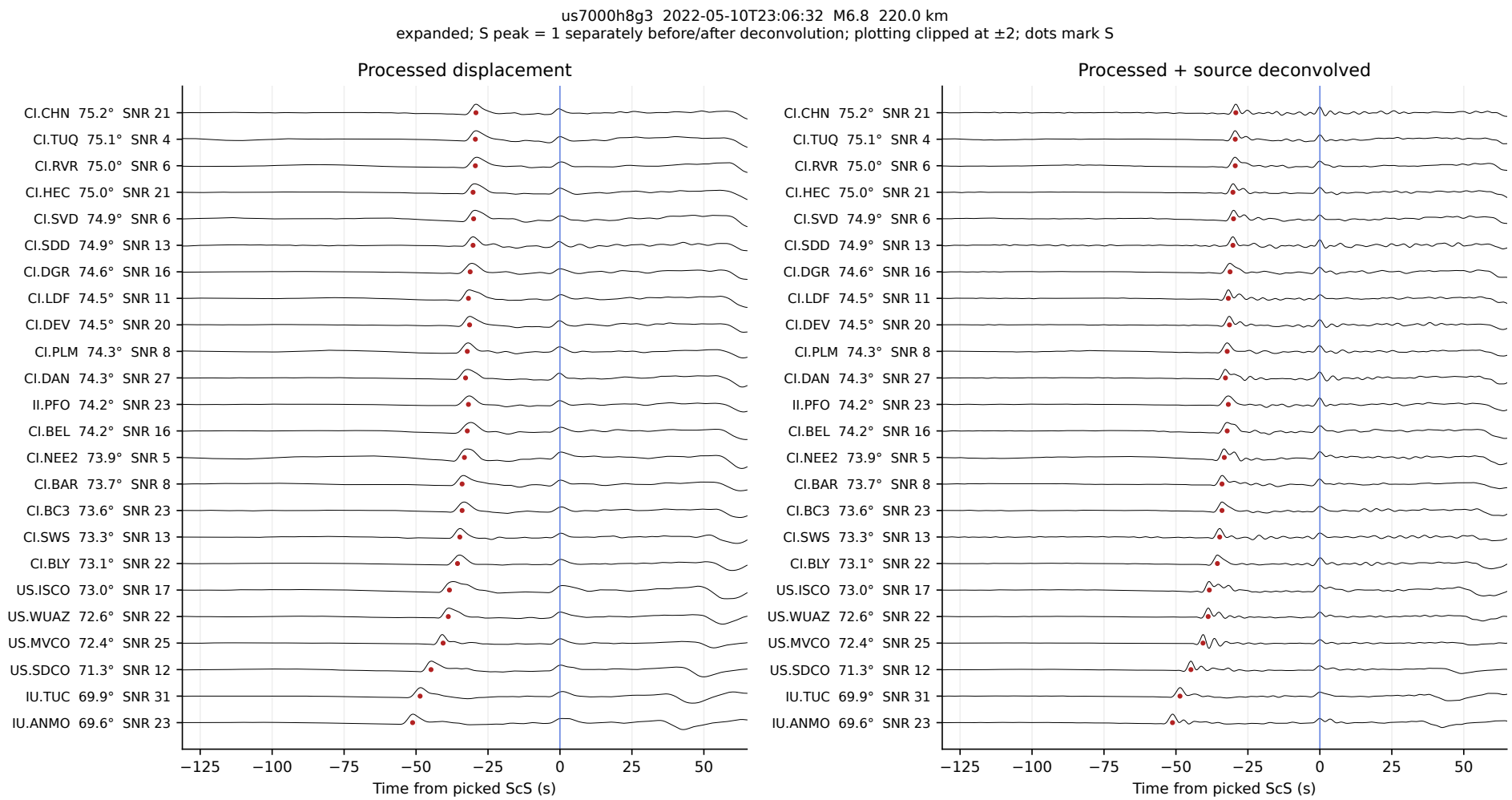
us70006qk4: 2 used traces; page 1/1



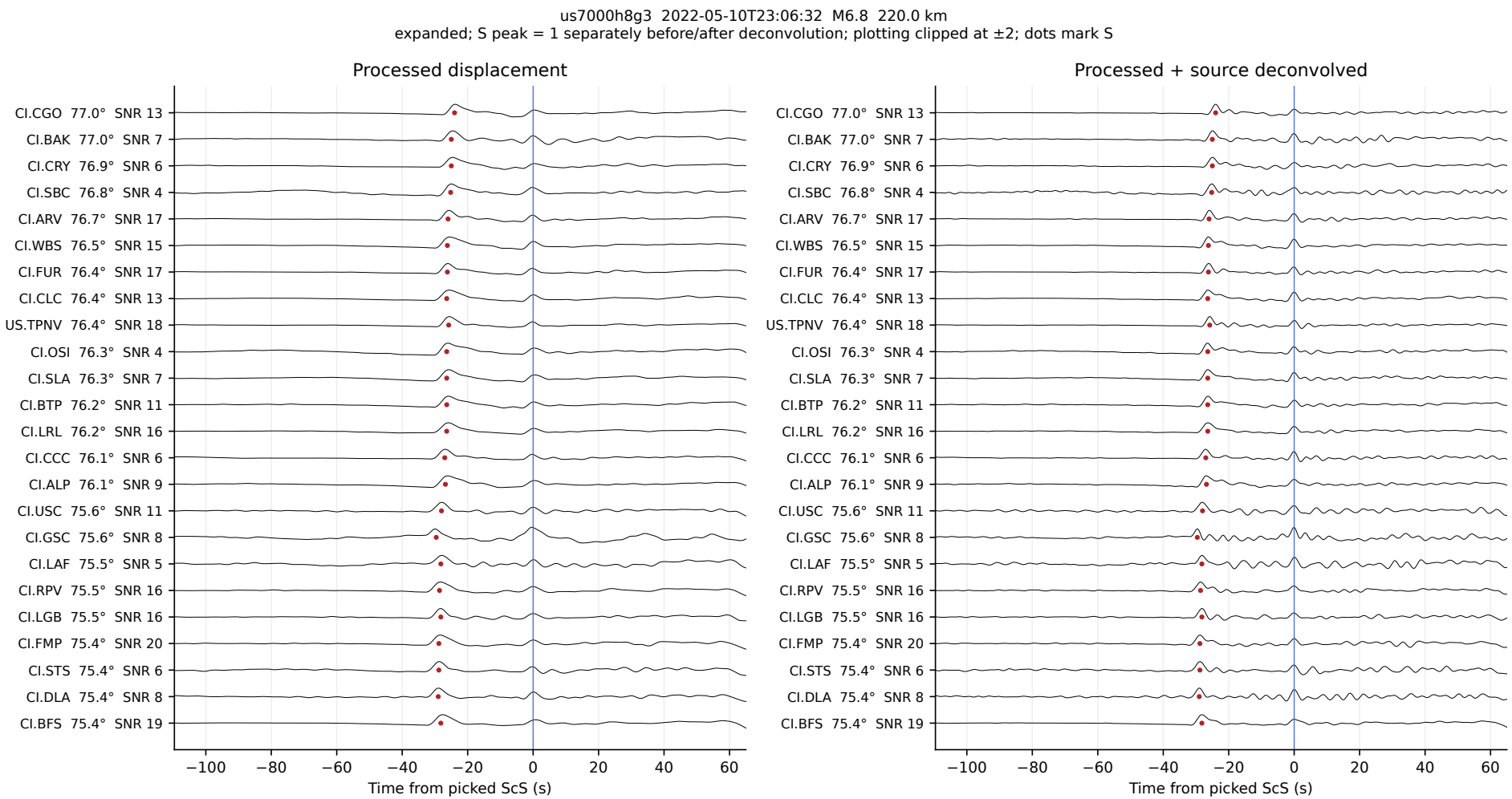
us7000gmw3: 20 used traces; page 1/1



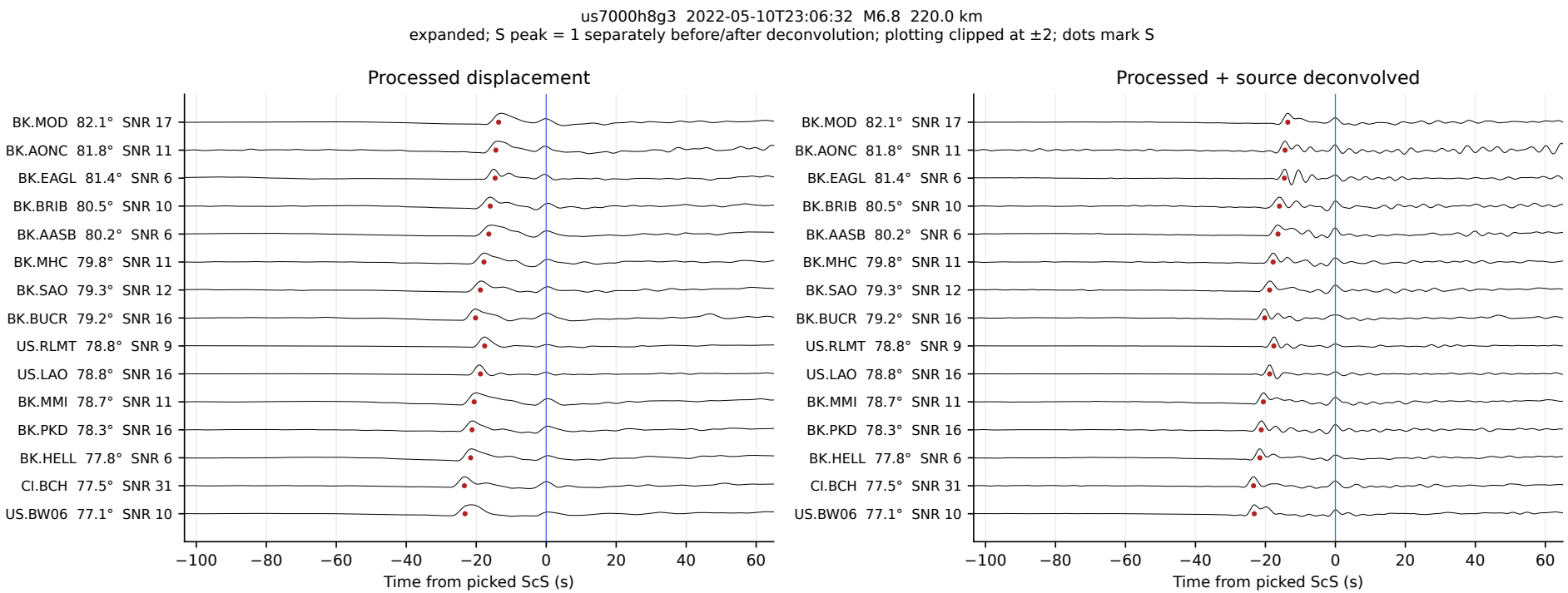
us7000h8g3: 63 used traces; page 1/3



us7000h8g3: 63 used traces; page 2/3



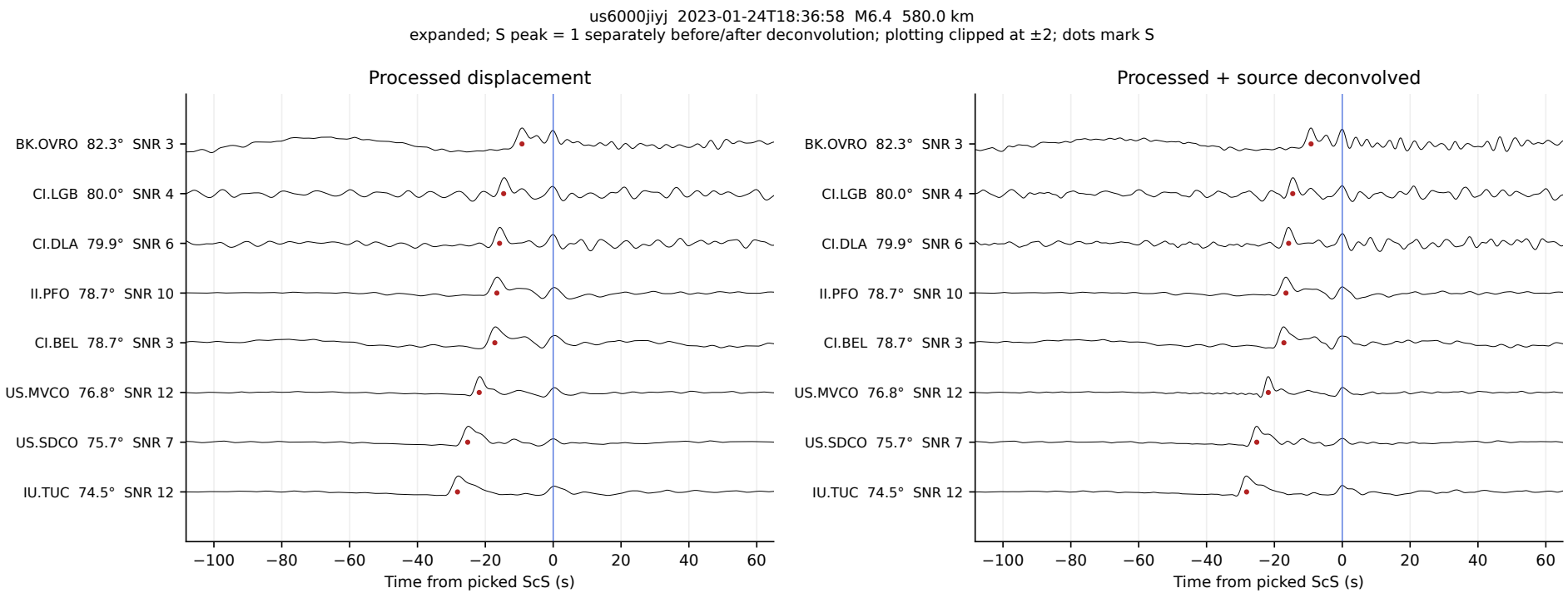
us7000h8g3: 63 used traces; page 3/3



us7000hcmn / 2022-05-26T12:02:21.110000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 95

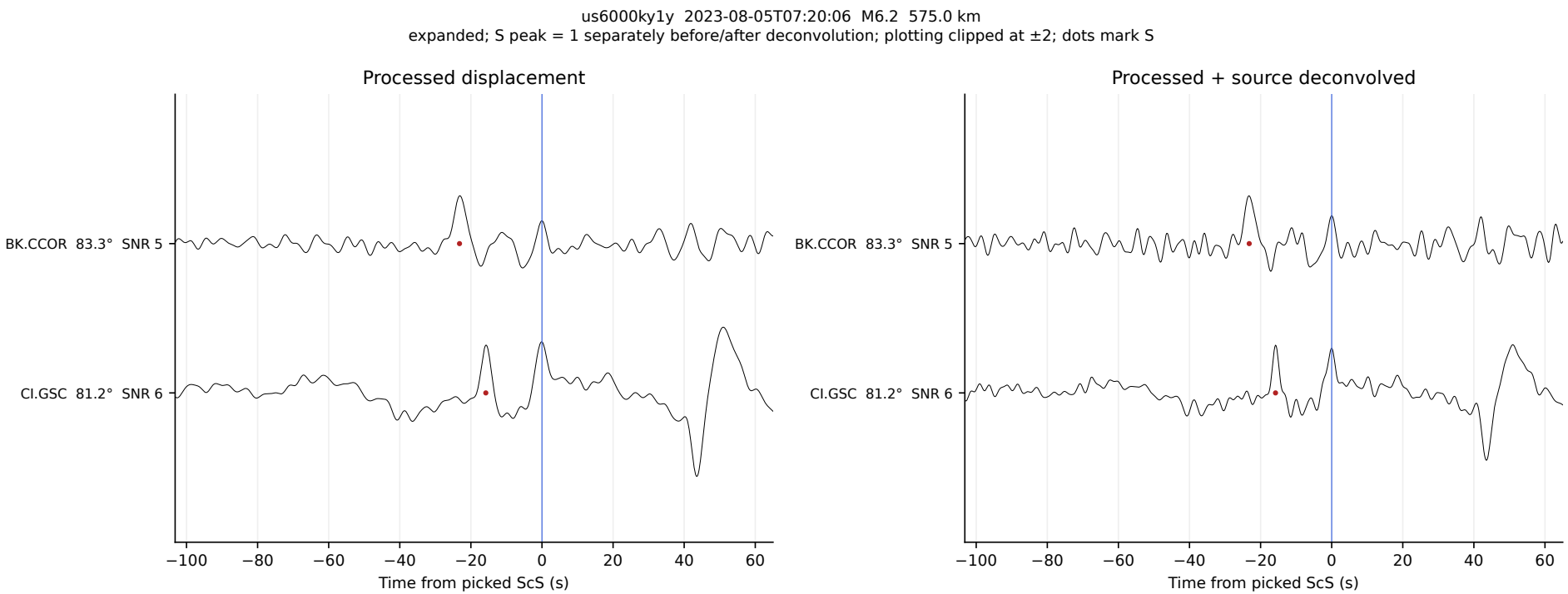
us7000j5x2 / 2023-01-20T22:09:39.053000Z
No accepted trace in this dataset.
candidate: 50
ScS peak at joint search boundary: 28
S or ScS below peak/RMS SNR threshold: 3

us6000jiyj: 8 used traces; page 1/1

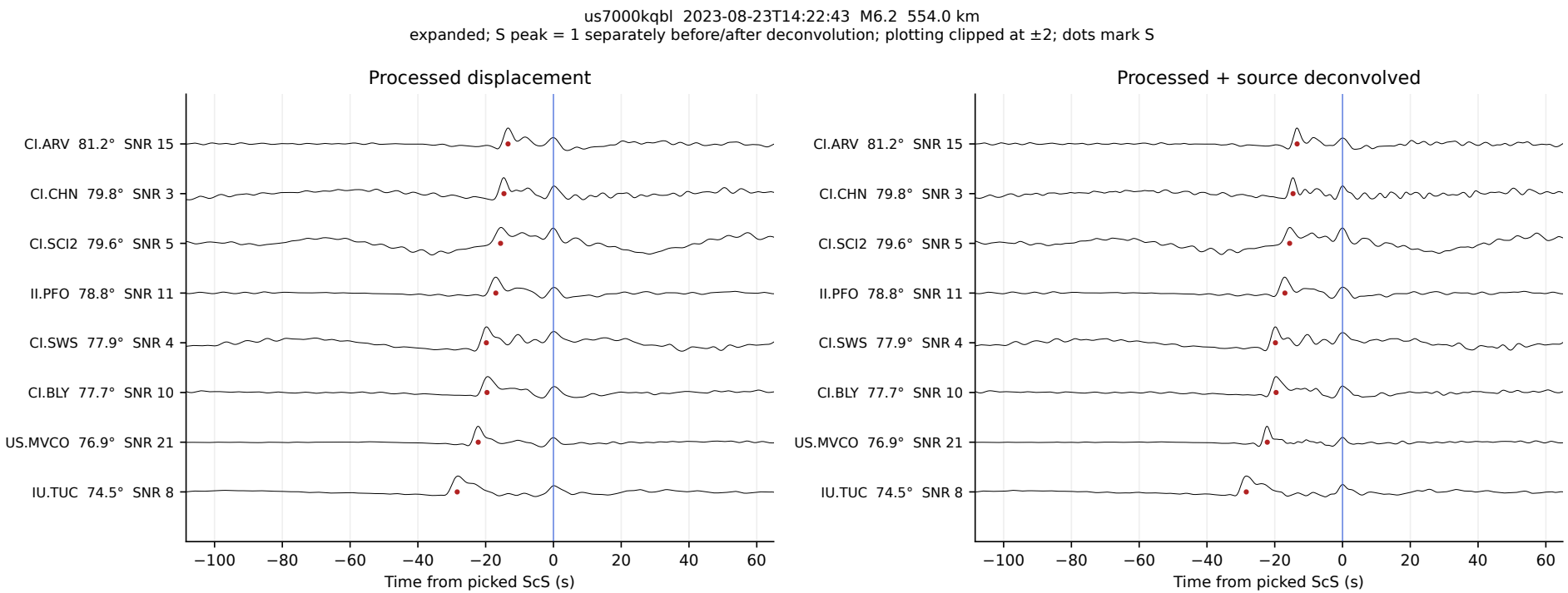


us7000jlxu / 2023-03-22T16:00:34.663000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 101

us6000ky1y: 2 used traces; page 1/1



us7000kqbl: 8 used traces; page 1/1



us6000legn / 2023-10-10T10:01:31.660000Z
No accepted trace in this dataset.
unstable or weak SH radiation (<0.6): 98

us6000m52p: 24 used traces; page 1/1

